

Review article

Comprehensive review of advanced methods for improving the parameters of machining steels

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ABSTRACT

The extensive range of steel types and classifications results in a diverse set of properties such as plasticity, heat resistance, strength, and corrosion resistance. This diversity enables the creation of alloys with varying machinability, from easy to hard-to-cut. This review focuses on the intricate interplay between mechanical characteristics of steel and machinability indicators such as tool wear, tool life, material removal rate (MRR), and chip formation. This review also highlights innovative strategies to enhance steel's machinability through the manipulation of cutting conditions. These approaches range from dry cutting to sophisticated cooling systems such as minimum quantity lubrication (MQL), cryogenic lubrication (CL), and high-pressure cooling (HPC). Additionally, the impact of tool design, including geometry and surface texturing, is scrutinized, along with the deployment of advanced tool materials and coatings. Furthermore, the paper investigates the efficacy of assisted machining techniques, including vibrational, thermal, and hybrid methods. A comprehensive review of advanced machining operations (encompassing turning, drilling, milling, and grinding) elucidates the potential improvements in surface integrity and machinability of steel. The review also addresses the challenges and future opportunities in this domain. In conclusion, the exploration into the machinability of steel underscores its significance as a versatile material in various industries. By understanding and optimizing the machining dynamics and employing innovative strategies, there is potential to further enhance the properties and machinability of steel alloys. This review provides valuable data compilation for researchers, engineers, and industry professionals seeking to advance the utilization of a large array of steel variants in manufacturing processes.

1. Introduction

The choice of materials is crucial in engineering design, significantly impacting performance, reliability, and cost-effectiveness [1,2]. Steels are alloys where the composition must contain two iron-carbon (Fe-C) elements, in which the mass fraction of carbon theoretically does not exceed 2.14 %, although in practice the concentration of carbon is not >1.5 % [3]. In addition to carbon, steel contains permanent impurities: silicon, manganese, sulfur, phosphorus, and other chemical elements [4]. The two formers (Si and Mn) are considered “good” impurities

because they may enhance strength, while the two latter (S and P) are the “bad” impurities, because they reduce the material's properties, although may improve machinability. Steel grades can generally be classified according to chemical composition (carbon and alloyed) or purpose (structural, instrumental, and special steels) [5], being applied in a wide range of industries including aerospace, automotive, power engineering, shipbuilding, tool production, medical instruments, etc.

In terms of machinability, steels are understood at various cutting parameters to achieve technological indicators such as the quality of the machined surface, cutting tool life, cutting force or power, and chip

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shapes [6]. Steels of different grades have very different physical and mechanical properties and therefore differ in machinability [7]. Low-carbon steels, for instance, tend to present burrs formation and chip sticking in the cutting tool at low cutting speeds [8]. In contrast, when machining carbon steels at high cutting speeds, material build-up does not occur, however, wear and cutting temperature increase, but the surface quality is higher [9]. When machining high-carbon steels, wear intensity and temperature increase even more [10]. Moreover, stainless steels are classified as hard-to-cut alloys due to their low thermal conductivity, high strength, and the tendency for chip sticking [11].

Therefore, when machining alloy steels, it is often necessary to use different tool materials and coatings, as well as finding the best cooling method, etc. [12], since their machinability varies from easily machined to hard-to-cut alloys [13]. The integrity of surfaces is a set of parameters that determine their surface micro geometry and subsurface modifications, as well as the nature and level of residual stresses that affect, among other things, the product, and its performance characteristics [14]. Therefore, it is important to consider the most advanced methods to improve machinability and surface integrity for different steel groups.

In the field of engineering and manufacturing, the machinability of steels is a topic that has many good review contributions, as highlighted in Table 1. These reviews often dissect the machinability of different steel grades, albeit typically through a narrow lens—focusing on specific techniques or objectives and frequently covering a range of materials or particular steel grades such as hardened steels. This observation underscores a significant gap: the absence of comprehensive reviews that simultaneously address the machining of steel across most grades and incorporate the main techniques for enhancing their machinability.

To bridge this gap, this review presents an extensive examination that traverses various machining processes, including turning, drilling, milling, and grinding. This effort aims to elevate machining performance across several steel grades through cutting-edge techniques while also spotlighting the challenges and future directions in this domain. The discussion extends to the impact of numerous factors on machining outcomes—cutting parameters, the cutting environment, tool geometry, workpiece material, tool coatings and textures, vibrations, and both thermal and hybrid assisted machining methods. Notably, it delves into the crucial balance between material removal rates and the preservation of workpiece surface integrity or cutting tool longevity, a pivotal consideration in steel machinability.

Furthermore, this review goes beyond existing literature by synthesizing advancements across a spectrum of cutting parameters, environments, tooling strategies, and assisted methods. It aims to highlight overarching trends, pinpoint critical research voids—such as the imperative for generalized machinability models—and juxtapose techniques across diverse output parameters like forces, temperature, and surface quality. This holistic analysis is poised to unveil information and opportunities heretofore unexplored, thereby enriching our collective grasp of machining processes and their practical applications.

This review distinguishes itself by examining current challenges and integrating recent advancements to elucidate the interplay among various machining techniques. By transcending a mere catalog of individual methods, the analysis addresses existing challenges while casting an eye towards future research avenues. This approach ensures that the review not only fills a notable gap in the scholarly discourse but also establishes a new benchmark for subsequent inquiries into the multifaceted nature of machining steels. In doing so, this work aspires to contribute profoundly to both academic research and industrial practices concerning the improvement of steel machining processes.

This review is structured by this introduction showing the relevance and purpose of the article, the Section 2 where the classification and properties of various steel grades is presented, and the Section 3 where the machinability of steels is discussed in terms of various cutting processes. In Section 4, several techniques and strategies for improving the machinability of different steel grades are discussed. The challenges and futures trends regarding machinability of steels and the article's

conclusion are shown in Sections 5 and 6, respectively.

2. Classification and properties of steels

Metallic alloys can be classified into two basic groups: ferrous and non-ferrous materials. Among ferrous materials, steels stand as the second group (the first one is cast iron) and have the superiority of carrying a combination of very important features such as formability and durability, good yield and tensile strength, and good thermal conductivity. They are preferred in numerous places in daily life that extends to transportation, vehicles, marine engineering, medical, construction, biomedical, aerospace, and aeronautics. Mainly, common steel is an alloy constructed of iron and carbon elements in which the content of carbon ranges up to 2 %. However, by adding several types of elements (in this case forming the alloy steels) and changing the carbon content, different microstructures and physical properties of steels can be achieved. The main types of steels and their subclasses are outlined in Fig. 1.

Common carbon steels (or plain carbon steels) have three main groups having different microstructures such as low carbon steels (pearlite, ferrite), medium carbon steels (pearlite, ferrite, martensite), and high carbon steels (pearlite, martensite). Low carbon steels contain small amounts of carbon (up to 0.3 %) and manganese ranging from 0.4 to 1.5 % which is generally produced as wires and flat shaped such as sheet metals [43]. AISI 1008, AISI 1010, and AISI 1020 are examples of low carbon steels encountered in the machining field. The acronym AISI stands for American Iron and Steel Institute. Such steels normally have low strength and hardness, high toughness, and ductility, and are not heat-treatable.

Some of the medium carbon steels broadly used in industrial applications can be sorted as AISI 1040, AISI 1045, AISI 1050, etc. Such materials are preferred for the manufacturing of railway vehicle parts, crankshafts, and gears [44]. They have greater strength and hardness and less toughness and ductility in comparison to low carbon steels. Medium carbon steels possess carbon content in the range of 0.3–0.6 %, along with manganese content between 0.6 and 0.65 %. They are suitable for quenching and tempering heat treatments, although need to be performed with high cooling rates and in thin sections to be effective.

High carbon steels are widely applied in prominent sectors and find application areas for railroad components, landing gears, stamping tools, dies, and punches [45]. Such steels cover the properties of high hardness and toughness along with low ductility and have resistance to brittle fracture. They present carbon content between 0.61 and 1.5 % and are usually applied after heat treatment (quenching and tempering). AISI 1060 and AISI 1080 are examples of high carbon steels.

The group known as “Alloy steels” is a general description made for specifying steels alloyed by several elements including some amounts ranging between 1 and 50 %. They contain specific amounts of alloying elements apart from those normally used in plain carbon steels, which promote changes in properties that allow the material to perform specific functions. Alloy steels are usually named according to the predominant element(s), such as nickel steel, chromium steel, chromium-vanadium steel, etc. Additionally, the term “alloy steel” can refer to any steel that contains minimum amounts of boron, chromium (up to 3.99 %), cobalt, niobium, molybdenum, nickel, titanium, tungsten, vanadium, zirconium, or any other alloying element added to improve mechanical properties and toughness after heat treatments. Thus, alloy steels include a wide variety of steels with different properties and applications including civil construction, shipbuilding, oil, automotive, and aeronautical industries.

Stainless steel is a type of alloy steel that is highly resistant to corrosion, rust, and staining. It is composed primarily of iron, with a minimum of 10.5 % chromium content by mass which forms a thin passive layer of chromium oxide on the surface, thereby acting as a protective barrier to various environmental conditions. Stainless steels have five main classes: ferritic, austenitic, martensitic, duplex (ferritic-

Table 1

Review articles that cover the machinability of steel to some extent.

Reference	Year	Review title	Number of references	Comments
[15]	2015	Machining of hardened steel — Experimental investigations, performance modeling and cooling techniques: A review	140	Focus on hardened steels. Effect of the workpiece's hardness, cutting parameters, tool coating, and cooling-lubrication conditions on machining performance are presented.
[16]	2017	Dry machining: A step towards sustainable machining e Challenges and future directions	120	Focus on dry machining. Research from 2000 to 2015 discussing the advantages and disadvantages of dry cutting is shown, as well as techniques that contribute to allowing its application. Different metals addressed; steels included. Various machining processes: turning, milling, drilling, and grinding.
[17]	2017	Material microstructure affected machining: a review	80	Review of microstructure alterations in machining processes discussing its effect on cutting forces and surface integrity.
[18]	2017	Potential of alternative lubrication strategies for metal cutting processes: a review	176	Review focusing on the sustainability of different cutting fluids presenting more sustainable cooling-lubrication strategies for machining various materials using different processes.
[19]	2017	A review identifying the effectiveness of minimum quantity lubrication (MQL) during conventional machining	182	Focus on the MQL technique and its advancements. Different machining processes are covered. Steels, aluminum alloys and difficult-to-cut materials are addressed.
[20]	2019	Role of textured tool in improving machining performance: A review	155	Review focusing on textured tools presenting different techniques and the texture's influence on the process. Steels, aluminum, titanium, and nickel alloys are addressed.
[21]	2019	An overview on economic machining of hardened steels by hard turning and its process variables	39	Focus on economic aspects in the turning of hardened steels in comparison to grinding. The influence of different input variables is discussed.
[22]	2020	A systematic literature review on machine tool energy consumption	238	Focus on energy consumption, especially in turning and milling. Different materials are addressed, steels included.
[23]	2020	Green Metalworking Fluids for sustainable machining applications: A review	135	Focus on sustainable cutting fluids applied in various machining processes. Different workpiece materials are addressed, steels included.
[24]	2020	Sustainable techniques in grinding: State of the art review	120	Review focusing on techniques to improve sustainability in grinding. Different materials are addressed, including steels.
[25]	2020	A review on recent development of minimum quantity lubrication for sustainable machining	176	Focus on the MQL technique and several strategies to improve it. Various materials are addressed, steels included. Different machining processes and various output parameters are discussed.
[26]	2020	A review of the chip breaking methods for continuous chips in turning	92	Review of chip break methods in the turning process. Different materials are addressed, including carbon steel, tempered steel, and stainless steel.
[27]	2020	A review of surface integrity in machining of hardened steels	82	Review focusing on the surface integrity of machined hardened steels. The effect of several input variables on different output parameters is discussed.
[28]	2021	State of the art of tool texturing in machining	135	Review of cutting tool texturing, presenting results on the machining of different materials, steels included.
[29]	2021	Advancements in material removal mechanism and surface integrity of high speed metal cutting: A review	232	Review focusing on material removal rate and surface integrity in high speed metal cutting (HSMC). Different materials are addressed, steels included. Various output parameters are discussed.
[30]	2021	A comprehensive review on research developments of vegetable-oil based cutting fluids for sustainable machining challenges	217	Comprehensive review focusing on vegetable cutting fluids. Machinability of steels and other materials are presented.
[31]	2021	A review on nanofluids in minimum quantity lubrication machining	147	Review focusing on nanofluids applied via MQL technique. Different machining processes and workpiece materials are addressed.
[32]	2021	Recent advancements in nano-lubrication strategies for machining processes considering their health and environmental impacts	250	Review of nano-lubrication strategies applied to different machining processes. Various workpiece materials are addressed: steel, aluminum, titanium, and nickel alloys.
[33]	2021	Progressing towards Sustainable Machining of Steels: A Detailed Review	207	Focus on sustainability of machining steels. Different cooling-lubrication strategies are discussed.
[34]	2022	A critical review on the progress of MQL in machining hardened steels	106	Focus on machining of hardened steels using the MQL technique. Different machining processes are addressed, as well as strategies to enhance the MQL technique.
[35]	2022	Recent progress and evolution of coolant usages in conventional machining methods: a comprehensive review	185	A comprehensive review focusing on the cutting fluid application in turning, milling, drilling, and grinding. Several types of metals are covered, steels included.
[36]	2023	State-of-the-art on minimum quantity lubrication in green machining	112	Review of the minimum quantity lubrication (MQL) technique. Different machining processes and workpiece materials are addressed.
[37]	2023	A review of the research on the variation of tool's motion trajectory and its influence on the formation mechanism of surface quality in ultrasonic vibration machining	213	Focus on ultrasonic vibration machining (UVM). Different machining processes and workpiece materials are addressed.
[38]	2023	Optimization techniques for energy efficiency in machining processes — a review	218	Review of optimization techniques to improve energy efficiency in turning, milling, and drilling. Different cooling-lubrication techniques and various output parameters are addressed. The review covers steels, aluminum, and difficult-to-cut materials.
[39]	2023	Research status of the influence of machining processes and surface modification technology on the surface integrity of bearing steel materials	132	Review focusing on surface integrity of bearing steels. Different manufacturing processes are discussed, as well as surface modification technologies for bearing steels.

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Table 1 (continued)

Reference	Year	Review title	Number of references	Comments
[40]	2023	A review of low-temperature plasma-assisted machining: from mechanism to application	125	Focus on low-temperature plasma-assisted machining. Different materials are addressed, steels included.
[41]	2023	Equipment–process–strategy integration for sustainable machining: a review	212	Review of sustainability of machining processes in terms of machine-tool, cutting tools, and cooling-lubrication conditions. Different machining processes are addressed, as well as various workpiece materials.
[42]	2023	Grey Relational Analysis-Based Optimization of Machining Processes: A Comprehensive Review	247	Focus on Grey Relational Analysis (GRA) optimization. Different machining processes addressed, including non-conventional machining. Various workpiece materials included.

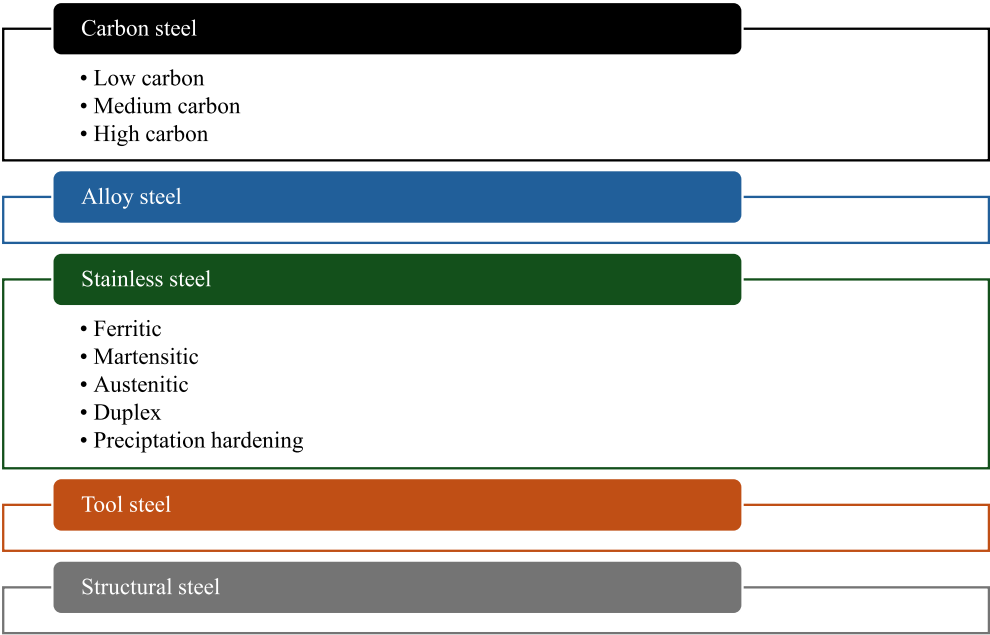


Fig. 1. Types of different steels mostly encountered in the literature.

austenitic), and precipitation hardening (PH). Each class presents specific features that are required in different applications. Ferritic stainless steels, for instance, are proper to integrate into automotive parts and combustion systems with their features of toughness, ductility, and cracking resistance [46]. AISI 430, AISI 409, and AISI 439 are some examples of such steels.

Austenitic stainless steels are a specific type of stainless steel that exhibits a face-centered cubic (FCC) crystal structure known as austenite. They typically present high levels of chromium (16–26 %) and nickel (6–22 %) that provide excellent corrosion resistance, even in aggressive environments. They are highly resistant to oxidation, staining, and pitting corrosion, and exhibit good mechanical properties such as high ductility, formability, and toughness. They are non-magnetic, even after significant cold working or deformation, thereby being suitable for applications where magnetic interference is a concern such as in electronic equipment or certain medical devices. Other applications are food processing, chemical processing, pharmaceuticals, architecture, and household appliances. Due to their excellent corrosion resistance, hygienic properties, and aesthetic appeal, they are also used in piping, tanks, heat exchangers, kitchen utensils, and architectural structures. Low thermal conductivity and hardenability make the austenitic stainless steels problematic for machining and the popular materials AISI 316L and AISI 304 are types broadly used [47].

Martensitic stainless steels are available for quenching and tempering to achieve a hard structure [48]. Besides, such steels have good corrosion and/or oxidation resistance along with good strength under certain temperatures and the ability to keep creep resistance even

at elevated temperatures. AISI 410, AISI 420 AISI 440A are the most encountered martensitic stainless steels in the machining field.

Compared with other types of stainless steels, duplex stainless steels have higher corrosion resistance thanks to their high nickel, chromium, and molybdenum content [49]. The primary feature of such steels is that they present ferrite and austenite phases together. AISI 2205, AISI 2507, UNS S32760, and DIN EN 10088-1 are the most common duplex stainless steels (the acronyms UNS and DIN stand for Unified Numbering System and Deutsches Institut für Normung respectively).

Precipitation hardening (PH) stainless steels are attracting attention in the last years owing to their mechanical properties along with superior weldability and forgeability [50]. For PH sub-groups, martensitic, semi-austenitic, and austenitic types are the most important ones. 17-4 PH, 15-5 PH, 13-8 PH stainless steels are mostly representative types of such materials. Aerospace, nuclear energy oil and gas, and automotive industries often use these materials thanks to the listed properties [51].

AISI O1, ASSAB 88, AISI H13, HcHcR D2 (High Carbon High Chromium), and AISI D6 steels are the most commonly worked types of tool steels in the literature. Such steels carry specific characteristics which are widened from hardness to wear resistance and toughness to softening resistance, making them suitable for cutters, reamers, and bits used for cutting [52]. Heat treatments are key points for tool steels and are not easy to carry out [53]. Without the correct heat treatment, tool steels are like ordinary steels, and each type has the proper routine to guarantee the properties requirements [54].

Structural steels are broadly preferred due to their light and stronger structure. They have good weldability and extreme hardness and

strength mostly owing to their utilization area requires high protection. Such steels have higher carbon content than mild steels. A36, A572, Strenx steels, and ArmoX steels are the mostly seen types of structural steels in the field. The chemical compositions of the main types of steels are shown in Table 2.

3. Machining parameters of steels

Steels have long been the foremost engineering material, extensively used in production sectors, with ongoing and extensive research into their machinability. While many studies focus on cutting steels and their findings, research incorporating additional data from experimenting with other metals is crucial. Detailed research has provided information into how material properties, which can vary at microscopic and even atomic levels, influence machinability. Machinability is highly sensitive, with results often assuming homogeneous microstructure and isotropic properties, but actual variations in microstructure can lead to significant differences. Even slight variations in residual element content within standard limits can affect machining performance, making it critical, as steels from the same producer but different batches can yield different machining results.

3.1. Machinability of steels

Understanding a material’s machinability in a process is crucial for estimating cutting power and output variables, as well as correlating with tool wear to assess economic viability. Machinability evaluation considers various output variables such as tool chip contact length, surface integrity (e.g., roughness, microhardness, residual stress), cutting force, temperature, chip morphology, and tool life. Machining steels poses challenges in chip control and tool wear due to their characteristics. While a complete review of steel machinability literature is impractical due to its vast volume, efforts have been made to highlight key findings covering a wide range of steel types.

The authors Sivaiah et al. [67] evaluated the machinability of AISI 52100 bearing steel in a comparative study involving MQL, flooding, and dry cooling-lubrication conditions with textured tools. Only one type of texture was utilized, comprising linear grooves perpendicular to the chip flow. The texture dimensions are a width of 50 μm, a depth of 20 μm, and spaces between successive grooves of 40 μm. Results showed that the textured tool with MQL lubrication significantly decreased cutting temperature, tool wear, and workpiece surface roughness compared to other conditions. Reductions of 25 %, 40 %, and 42 %, respectively, were observed compared to the dry condition and the cutting tool without texture.

Pal et al. [68] evaluated the machinability of AISI 4340 hardened steel using Al₂O₃ + TiC mixed ceramic tools. Three variants of hardened steel, with hardness levels of 35, 45, and 55 HRC, underwent turning operations with varied feed rates (0.081 to 0.15 mm/rev), cutting speeds (100 to 200 m/min), and depths of cuts (0.1 to 0.5 mm). Results indicated that higher hardness led to increased forces and cutting temperature, along with improved workpiece surface roughness. Surface roughness showed a strong correlation with feed rate, while cutting temperature and forces correlated significantly with cutting speed and depth of cut, respectively.

Still dealing with turning, Das et al. [69] investigated the machinability of AISI 4340 steel applying four different types of nanofluids (ZnO, CuO, Fe₂O₃, and Al₂O₃) mixed in deionized water (at a concentration of 0.1 % by volume) applied by MQL (150 mL/h), and the properties studied in these fluids were thermal surface tension, conductivity, viscosity, and contact angle. Fe₂O₃ nanoparticles measured approximately 11 to 15 nm, while others ranged between 30 and 50 nm. Consequently, CuO nanofluid demonstrated the most favorable influence on the machinability of the steel, considering variables such as workpiece surface integrity, cutting forces, and tool wear, followed by ZnO, Fe₂O₃, and Al₂O₃ nanofluids. The lower viscosity of CuO

Table 2
Content and mechanical properties of several types of steels.

Ref.	Material	Alloying elements (wt%)										TS (MPa)	YS (MPa)						
		Fe	C	Si	Mn	Ni	Cr	Mo	V	S	Cu			P	Co	W	N	Nb	Ti
[55]	Low carbon steel AISI 1020	Bal.	0.190	0.19	0.98	-	-	-	-	0.031	-	0.016	-	-	-	-	-	480	270
[56]	Medium carbon steel AISI 1045	Bal.	0.46	0.24	0.45	0.01	0.015	-	-	0.003	-	0.008	0.01	-	-	-	-	729	507
[57]	High carbon steel AISI 1080	Bal.	0.79	0.33	0.73	0.07	0.10	0.03	-	0.005	-	0.023	-	-	-	-	-	740	358
[58]	Alloy steel 27MnCr5	Bal.	0.266	0.234	1.193	0.106	1.080	-	-	0.0313	0.109	-	-	-	-	-	-	590	369
[59]	Ferritic stainless steel AISI 430	Bal.	0.035	0.31	0.34	0.14	16.01	-	-	0.0011	-	0.017	-	-	0.042	-	-	530	295
[60]	Martensitic stainless steel AISI 410	Bal.	0.12	0.39	0.39	0.09	13.12	0.006	0.09	<0.001	0.121	0.026	-	-	-	-	-	1270	1010
[61]	Austenitic stainless steel AISI 304	Bal.	<0.08	<0.75	<2	8-10.5	18-20	-	-	<0.03	-	<0.045	-	-	<0.1	-	-	515	205
[62,63]	Super duplex stainless steel UNS S3275	Bal.	0.03	2.06	1.20	6.1	24.6	4	-	0.02	-	0.03	-	-	0.25	-	-	720	473
[64]	Precipitation hardening steel 15-5PH	Bal.	0.029	0.32	0.66	4.4	15.3	-	-	0.001	3.38	0.024	-	-	-	0.32	-	1191	663
[65]	Tool steel AISI D6	Bal.	2	0.3	0.75	-	12.5	-	-	-	-	-	-	1.25	-	-	-	702	529
[66]	Structural steel ASTM A36	Bal.	0.15	0.18	0.69	0.088	0.15	0.0195	0.0048	0.0040	0.24	0.018	-	-	-	0.0024	0.0012	400-550	250

nanofluids, enabling better penetration into the cutting area, along with their higher heat transfer capacity, resulting in a lower cutting temperature, likely contributed to this outcome.

In the work developed by Shao et al. [70], the studied process involved milling stainless-steel workpieces with 3 % Co and 12 % Cr, a highly heat-resistant material crucial for power plant critical parts. Its machining is deemed challenging. To gauge machinability, tool wear and failure mechanisms were examined for carbide tools coated with TiCN/TiN, featuring different rake angles (17° and 28°). A longer tool life was noted with a smaller rake angle, albeit accompanied by increased vibration and inferior surface finish. In machining this steel, tool wear progressed through three stages: abrasive wear, adhesion, and diffusion, culminating in tool failure primarily due to breakage/chipping.

Various studies in this section explored the machinability of different steels using diverse cutting techniques. Findings revealed that textured tools with Minimum Quantity Lubrication (MQL) reduced cutting temperature, tool wear, and surface roughness. Steel hardness affected cutting forces and temperature, while CuO nanofluids showed benefits for machinability. Tool rake angles influenced tool life and surface finish in stainless steel machining, with different wear mechanisms observed. Optimal parameters were identified for achieving desired surface roughness and minimizing cutting wear in D2 steel. Overall, these studies contribute to understanding and optimizing steel machining processes based on specific material properties and cutting conditions.

3.2. Surface integrity

The machined surface's final condition is influenced by various phenomena such as material distortion, breakage, elasticity recovery, thermal expansion, oscillation, and enduring tensions. These factors can have diverse effects on the machined surface, making surface integrity crucial. Surface integrity refers to the overall quality of a surface, encompassing the alterations it undergoes. Therefore, surface integrity isn't limited to surface texture or geometric shape but also includes other surface characteristics in service and layers beneath it [71]. Among the main characteristics of surface integrity, the following stand out.

3.2.1. Surface topography

Roughness can be defined as composed of small irregularities (micro geometric errors) resulting from the intrinsic action of the machining process, and, usually, with ripples and flaws that must be avoided on the machined surface, as they represent manufacturing errors [72]. In this way, the roughness is an output variable specified according to the application of the workpiece machined. A wide range of applications of machined steel necessitates a diverse array of surface quality requirements.

In this theme, Leksycki et al. [73] researched the turning of 17-4 PH stainless steel varying the lubrication conditions, feed rate, and cutting speed. Topographic parameters, including roughness and contour maps, were analyzed using a profilometer, correlating these variables with input parameters during cutting. Compared to dry machining, near-dry conditions (MQL) resulted in a 48 to 38 % decrease in surface roughness (Sa). For the Sa range of 0.4–0.8 µm in wet and dry cutting atmospheres, isotropy was 20 %, while in near-dry tests, it reached 60 %.

Li et al. [74], using cemented carbide tools, researched the machining of AISI 304 austenitic stainless steel, characterized by its low machinability, which wears out the cutting tool rapidly and makes it difficult to obtain a good roughness. Parameters aimed at optimizing machined part topography (via cylindrical and face turning) were assessed. Findings revealed that increasing the feed rate or cutting speed led to higher tool wear (crater, flank, and notch), increased work material adhesiveness (Built-Up Edge - BUE and Built-Up Layer - BUL), and simultaneous changes in workpiece roughness. However, surface imperfections (e.g., scratches, feed marks, adhered material) were

randomly created and dispersed. Optimal cutting parameters for this metal were determined as a feed rate of 0.18 mm/rev and a cutting speed of 120 m/min.

Szczotkarz et al. [75] evaluated AISI 1045 steel topography in turning process with carbide tools in different coatings: TiC, TiAlN, and TiN. The authors experimented with various feed rates (0.05–0.250 mm/rev) and cutting speeds (125–325 m/min). Topography was assessed using the focal method, providing 3D characteristics, visual representations of contours, isometric perspectives, and curves illustrating material ratios. Findings showed that the lowest roughness was achieved with higher feeds and lower speeds with TiAlN coating. Conversely, with TiC, higher speeds correlated with lower roughness, along with more uniform valleys under these conditions.

Maruda et al. [76] evaluated the topographic effects on AISI 1045 carbon steel by varying the lubrication conditions of the turning process. The discussed research focused on evaluating surface roughness of machined steel using different machining parameters and lubrication conditions. Various methods were tested, including dry, MQL, cooled MQL, and cooled MQL with a phosphate ester additive. Results showed that the latter method resulted in the lowest values of Sq on the machined surface and lower machining temperatures due to a more uniformly formed tribofilm. This tribofilm prevented volumetric wear on the tool, leading to improved workpiece roughness. In summary, optimizing cutting parameters and lubrication significantly impacted surface roughness, enhancing the quality of the machined part and reducing tool wear.

Mahapatra et al. [77] investigated the use of multi-walled carbon nanotubes mixed nanofluid with minimum quantity lubrication for hard turning of hot work AISI H13 steel. The study explores various machining parameters' effects on tool vibration and surface roughness. Findings suggest that new-generation AlTiSiN coated carbide tools are efficient for machining hot work tool steel with environmentally conscious minimum quantity cooling-lubrication.

3.2.2. Microstructural alterations

Metallurgical alterations in the metal's microstructure resulting from conventional machining processes primarily involve phase transformation, recrystallization, grain fragmentation, and material refinement. These changes are more frequent under severe cutting conditions or with worn tools. The mechanical (plastic deformation) and thermal effects during machining can induce martensite formation within austenitic stainless steel, diminishing its antioxidative properties. Mechanisms of martensitic transformation encompass spontaneous, stress-assisted, and strain-induced transformations. [78].

Regarding this phenomenon in austenitic stainless steels, Queiroga et al. [79] studied how changes in the microstructure of these metals can be related to hydrogen embrittlement effects. Turning tests were conducted on steel surfaces with strain-induced martensite, exposed to hydrogen permeation via cathodic charging. The study found that AISI 304 stainless steel experienced significant embrittlement, whereas AISI 310 steel did not exhibit this effect. This underscores the importance of controlling machining conditions, particularly for austenitic stainless steels.

In the study of Li et al. [80], a developed multi-physics model is employed to numerically simulate the evolution of microstructure in the machined subsurface of H13 steel. In this simulation, cutting parameters were adjusted for milling. Experiments demonstrated that increasing the cutting parameters resulted in the grain size of the upper machined surface varying around 330 nm and gradually expanding within the subsurface until reaching its initial dimensions of 20,000 nm. The influenced depth fluctuated between 25 and 45 µm. The formation mechanism of cell structure and displacement (grain refinement) stemmed from elevated deformation in steel machining and its martensitic structure.

Still dealing with milling, Li et al. [81] investigated the modification and transformation of the AISI H13 steel microstructure. To characterize

this alteration, several techniques were used: scanning and transmission electron microscopy, nano-indentation technique, and electron back-scattering diffraction. The milling results under different cutting parameters indicated that plastic deformations caused by machining loads increased the proportion involving grain boundaries with small angles (2° – 10°) alongside a decrease in local misorientation angle. There was an increase in the nano hardness of the material, reflecting grain refinement and dislocation accumulation. The machined surface layer exhibited significantly refined grains at the nanoscale, accompanied by a noticeable increase in dislocation density under various cutting parameters. The process of grain refinement is believed to involve the formation of cellular substructures originating from the accumulation of dislocations.

W. Zhang et al. [78] formulated an analytical model to foresee alterations in the microstructure of AISI 304 austenitic stainless steel in the turning process. This simulation modeled the kinetics of martensitic formation originated by strain. Experiments were carried out parallel to the simulation, whose values obtained corresponded to each other. The results indicated that a higher depth of cut allied with a lower cutting speed could result in a higher martensite content below the surface.

In summary, the metallurgical changes occurring in the microstructure of metals during conventional machining processes were discussed in this section. These changes involve phase transformation, recrystallization, grain fragmentation, and material refinement, particularly under severe cutting conditions or with worn tools. For austenitic stainless steels, the presence of martensite caused by plastic deformation and thermal effects can reduce the antioxidant effect and lead to hydrogen embrittlement effects, emphasizing the importance of controlling machining conditions. Studies on H13 steel and AISI 304 austenitic stainless-steel show that varying cutting parameters in milling can result in grain refinement, dislocation accumulation, and alterations in the microstructure, including the formation of martensite. Simulation and experiments in turning processes also support the correlation between cutting parameters and martensite content in the microstructure of AISI 304 stainless steel.

3.2.3. Mechanical properties

Mechanical alterations caused by conventional machining processes are mainly related to residual stress, microhardness, and work hardening on the workpiece surface (typical of rough machining). The residual stresses can be of tensile or compressive natures and of vary in intensity and depth. Normally, the residual stresses caused by thermal loads are tensile and those caused by mechanical loads are compressive. The compressive residual stresses are beneficial as they increase the fatigue strength of the part. Typically, the greater the tool wear, the more intense and profound the residual stresses [71].

In this context, F. Zhang et al. [82] investigated the residual stresses in the machining of AISI 52100 steel. The study included simulations and experiments for their validation, and the orthogonal cut was performed by a face turning on tubes with PCBN tools. In the white layer, the residual stress was compressive, and for low cutting speeds, it originated from plastic deformation. Since tensile residual stress is caused by thermal energy, it is normally produced during the cutting at high cutting speeds. Stresses of tensile nature are introduced in both the white and dark layers if the cut is performed by a worn tool.

Clavier et al. [83] assessed the correlation between tool wear and residual stresses in a 15-5 PH stainless steel workpiece. Unlike many previous works that solely relied on experimental stages to validate simulations, this study took a different approach. A numerical model was developed based on experimental data, enabling the researchers to establish a connection between the observed thermomechanical loadings and the resulting data from the turned surface. In the simulation, wear on the flank face and rake face, as well as adhesion, were tested. The results revealed that an increase in flank face wear had a notable impact on the residual stress of the workpiece, leading to an increase in the depth of the affected area. Wear resulting from adhesion exhibited

similar effects as flank face wear, albeit to a lesser extent. In contrast, the wear on the rake face did not show any significant influence on the residual stress.

The above text discussed the mechanical alterations caused by conventional machining processes, focusing on residual stresses, microhardness, and work hardening on the workpiece surface. Compressive residual stresses are beneficial as they increase the fatigue strength of the part, while tensile residual stresses are normally produced during cutting at high speeds. Studies on AISI 52100 steel and 15-5 PH stainless steel reveal the relationship between tool wear and residual stresses, with flank face wear having a notable impact on the workpiece's residual stress. Another study on the 42CrMo4 steel investigates the corrosion resistance related to mechanical properties, finding that compressive residual stresses contribute to enhanced corrosion resistance. Cutting speeds and feed rates also influence the magnitude of residual compressive stress on the machined surface.

3.2.4. Residual stresses

Residual stresses arise from non-uniform plastic deformation during machining operations due to thermal gradients and mechanical loads applied to the workpiece material. They can exist in either compressive or tensile form within the subsurface layers and affect the functional properties and service life of machined parts [84]. Compressive residual stresses are generally considered beneficial as they increase the fatigue strength and resistance to crack initiation in steels. However, tensile residual stresses have the opposite effect and can reduce the fatigue endurance of components [85].

The type and magnitude of residual stresses are influenced by various machining parameters such as cutting speed, feed rate, depth of cut, tool geometry, tool wear condition, and cooling method employed [86]. Higher speeds and feeds tend to increase tensile residual stresses due to greater frictional heating at the tool-workpiece interface [87]. On the other hand, cryogenic cooling and minimum quantity lubrication techniques have been shown to promote more favorable compressive residual stresses through effective temperature control [88]. Proper selection of these process variables is therefore important from the standpoint of residual stress optimization and machinability.

Damir et al. 2024 [89] studied the application of cryogenic machining (CM) using Liquid Nitrogen (LN2) for improving the machining performance of tool steel H13. Experimental tests were conducted for turning and milling operations on H13 steel, comparing cryogenic cooling with flood coolant. The findings revealed that cryogenic machining led to reduced cutting forces, improved surface roughness, and extended tool life. Moreover, the use of cryogenic cooling induced desirable compressive residual stresses on the surface of the workpiece, essential for the longevity of the mold in service. Overall, cryogenic machining demonstrated enhanced productivity, part integrity, and environmental performance compared to flood coolant.

Barcelos et al. 2024 [90] presented a study that analyzed residual stresses and surface integrity in the machining of low carbon manganese-alloyed steel after plasma arc cutting. The research found that plasma cutting generated compressive residual stresses on the workpiece beneath the cutting edge, extending to approximately 550 μm from the surface. The study also compared conventional and high-feed turning techniques, revealing that high-feed turning demonstrated superior wear resistance compared to conventional turning under the evaluated machining conditions.

Zhou et al. 2024 [91] focused on predicting residual stress in the orthogonal cutting of H13 steel using a hybrid modeling approach. It combines finite element analysis and neural networks to efficiently predict cutting residual stress, emphasizing its importance in evaluating surface integrity and die performance. The study explores the effects of tool structure parameters and cutting parameters on residual stress distribution in H13 steel, highlighting the impact of these factors on maximum compressive and tensile stresses. The findings demonstrate the potential of this method to provide a flexible and efficient model for

optimizing cutting conditions to control residual stress in H13 steel and other metals.

Residual stress characterization provides valuable data into how different machining strategies impact the functional integrity and performance of steels [92]. Nondestructive techniques like X-ray diffraction are commonly used to measure the near-surface residual stress distribution in machined parts [93]. The results can then be correlated with machinability indicators like tool wear, surface roughness and dimensional accuracy, as well as fatigue strength [94]. This aids in identifying process settings that minimize tensile stresses and maximize compressive stresses to improve the machinability and service life of machined steel components [95].

3.3. Tool wear and tool life in the machining of steels

Tool wear depends on the cutting tool’s material and type of the machining operation, which determines the exposure conditions such as temperature, forces, vibration, and main dynamics of cutting mechanisms [96]. Zhao and Liu [97] state that there are also influential factors that can be applied in cutting inserts for improving the ability to cut, such as tool geometry and coating technologies-related properties i.e., material and method. The efforts were directed to improvements of the tool materials along with adopting new tool geometries and coating technologies in the last half century until the 2000s. After that moment, novel tool geometry approaches widened, and surveys on determining the optimal cutting parameters range found popularity with comprehensive artificial intelligence applications.

According to Jamil et al. [98], the most effective way to improve the machining tribology of modern materials is lubricating and cooling strategies. Another approach is the use of more advanced cutting tool materials. Since steel represents a raw material for the world’s most significant engineering and construction products that lengthens from transportation to surgical operations. Precision and reliability are the main demands during the production process of these special materials. Among various manufacturing technologies, machining stands in the last stage of a whole procedure that makes it critical as per the ability to determine the ultimate conditions of the product. In this perspective, cutting tool properties should be designated by the experts for controlling tool wear and extending tool life to create high-quality products.

While designing the machining of steels theoretically, researchers and operators take into account many factors in machine tool and

cutting tool aspects. The main cutting parameters optimization is the primary necessity that requires deep knowledge and experience according to the type of steel being machined. According to Gupta et al. [99], this promotes the chip-removing mechanism and changes further developments related to surface integrity, workpiece tolerance, etc. In addition, cutting tool parameters such as geometry, material, and coating strategy need to be criticized since there are numerous cutting tool manufacturers in the sector. Despite these companies offering the same products including these factors when looking from outside, it is hard to say that they will give the same performance. Therefore, it should be noted that small details in the application stages such as the holding time during coating, the amount of elements in the material, and a small curve on the accurate point of the tool body have dramatic impacts on the cutting ability.

Modern methods applied from the beginning of the millennium focused on the performance improvement of cutting tools that depends on employing several cooling approaches. Jamil et al. [100] indicated that the primary intent here is to propagate the mist oil or sub-zero liquid in the gas form into the chip-tool-workpiece interfaces for improving the tribological conditions. Another option according to Colantonio et al. [101] for improving the tool life is using artificial intelligence methods that depend on monitoring progressing wear index via signals acquired by sensors integrated around the cutting zone. The primary factors affecting the tool wear and improving the tool life are outlined in Fig. 2.

Aslan [102] demonstrated that flank wear induced by the abrasive wear mechanism is an indicator for tracking the tool life as it develops on the main cutting edge. Most of the studies evaluate the progressive flank wear to check the health of the cutting tool and its remaining useful lifetime. The fracture of the cutting insert can be observed when the cutting tool exposes mechanical fatigue, and milling-like intermittent operations trigger this type of wear. Hatt et al. [103] showed that crater wear develops on the rake face of the cutting tool induced by the diffusion mechanism that is mostly observed under high temperatures. Chipping is observed especially for hard-to-cut materials which cover alloys and hardened steels. Thermal fatigue causes shocks on the cutting insert and results in thermal cracks through the rake face of the tool, mostly common in intermittent cuts. Notching is another type of tool wear that occurs at the back of the main cutting edge as a result of an oxidation mechanism. All of these types of tool wear reduce the cutting performance in the next cycles of the operation and affect dramatically

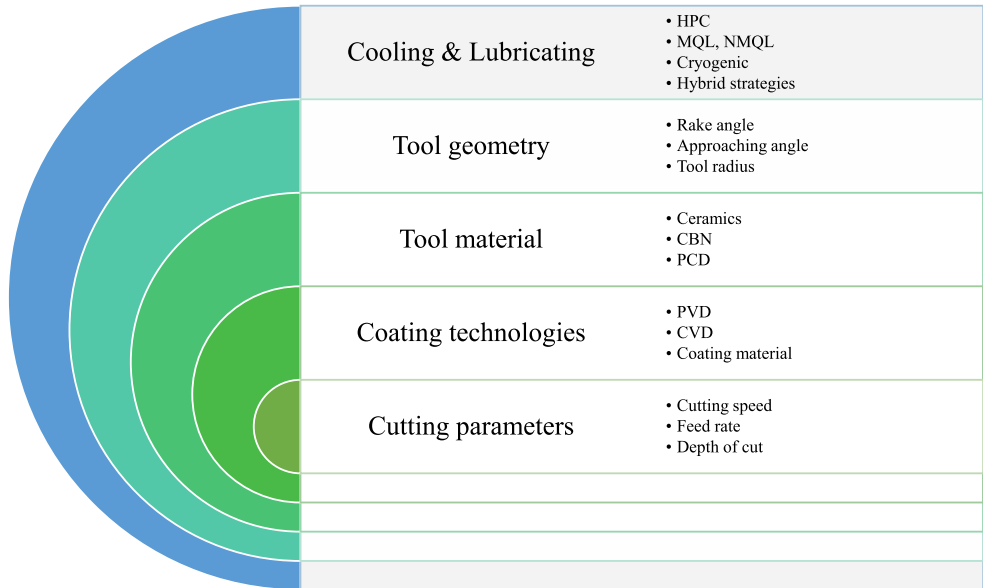


Fig. 2. Primary factors affecting tool wear and improving tool life.

the surface conditions.

3.4. Material removal rate (MRR)

Machining industries struggle with the minimization of energy consumption since such enterprises should optimize total outgoings. The primary motivation for this is improving efficiency which highly depends on reducing the cutting time. According to Kamble et al. [104], increasing cutting parameters provides higher material removal per unit time, thereby increasing MRR. However, since these variables affect entirely the cutting mechanism and play a role in machining characteristics, it is not possible to maximize them directly without looking at other machinability aspects. For example, an increase in feed rate and cutting depth promote an increasing impact on cutting force components, while cutting speed has the exact opposite effect in turning operations. Cutting forces themselves have a major influence on the wear behavior of the cutting tool, chatter stability, cutting temperatures, and surface integrity of the workpiece. On the other hand, the chip formation mechanism including chip shape, color, and removing capability needs to be considered when evaluating the success rate operation.

The influence of cutting parameters on the MRR and their side effects on other machining criteria are schematized in Fig. 3, where the minus signal refers to the uncertain effect of each parameter on the machining characteristic, and the up and down arrows show the increasing and decreasing trends for these variables, respectively. This addresses that many specifications need to be calculated in determining the MRR in metal cutting.

One important aspect when determining the MRR is deciding the operation type in terms of surface integrity, which can be rough, finish, semi-rough, precision, ultra-precision, and ultra-high-precision machining operation. This aspect was studied by Liao et al. [105], who found that the main approach that differentiates these operations is the depth of cut that need to be appointed for the expectation of surface quality. After the selection of the cutting depth value, cutting speed and feed rate are adjusted considering the material's properties. At this stage, maximization of these parameters is preferred but this approach causes excessive increases in cutting forces, reduced tool life, and surface roughness deterioration.

Therefore, the optimization of MRR is an important issue in the cutting of materials. Especially for metal cutting, determination of the ideal MRR is highly critical considering that these kinds of materials play a vital role in the production of daily life products. At this point, different types of metals need to be discussed for the final decision according to a given application. Surgical and medical products, for instance, require sensitively created surfaces, especially at fine profiles. The defense industry requires products that demand durable and high-

strength constructions along with finely manufactured surfaces.

3.5. Chip morphology

Pacella [106] stated that chip morphology is a crucial representative of healthy metal cutting since the contact conditions between the tool and removed material determine the basic dynamics of machining. Chip removal is a dedicated issue that can be accepted as an art along with an engineering problem. There are primary demands and expectations from chip removing and breaking operations such as i) removing the heat from the cutting area quickly, ii) manufacturing the desired surface by creating regular serrating that reflects the shape of the surface, iii) protecting the cutting tool from quick wear and elongates the tool life.

In some way, chip formation is related to shear stress distribution, material deformation, tribological conditions, and friction coefficient. Zhu et al. [107] showed that to determine such factors affecting the chip morphology, it is inevitable to establish accurate cutting geometry. Otherwise, spoiled chips reveal that will demolish the stability of machinability characteristics such as cutting forces, vibrational indicators, tool wear index, surface integrity, etc. demonstrated by Sales et al. [108]. Thus, the classification and characterization of the removed chips is of great importance. For this purpose, the advantages, and challenging sides of different types of chips need to be addressed.

Basically, there are three chip formation mechanisms in metal cutting operations according to Usca et al. [109]; i) self-breakage refers to the breakage that the chip will separate from the surface with no external effect, ii) breakage by crushing to the cutting tool, iii) breakage by crushing to the workpiece material. Among these processes, the first one is desirable due to its minimal effect on the surface morphology and microstructural features of the cutting tool and work material. The main types of metallic chips can be separated into groups that are represented in Fig. 4. Most importantly, long chips have a great potential to tangle the cutting tool holder and workpiece material, thereby entering the cutting zone and disturbing the material removal mechanism. This refers to the non-breakage of the chip during a remarkable period.

Şap et al. [110] stated that elongated chips are unable to eliminate the excessive heat around the cutting area which will increase friction between tool and chip and such a process causes fast wear of the tool. Moreover, chips longer than the desired measures would reduce the surface quality, especially for turning and drilling operations. Ideal chip length can be understood by the serrating texture and general view of the “c” shape. Oppositely, chips that have a length lower than these dimensions are hard to control and have a formation process as rupturing rather than cutting. They may be called short, extra small, or dust-type chips that can be faced during cutting hard-to-cut materials and composites. Since metallic materials cover a wide range of

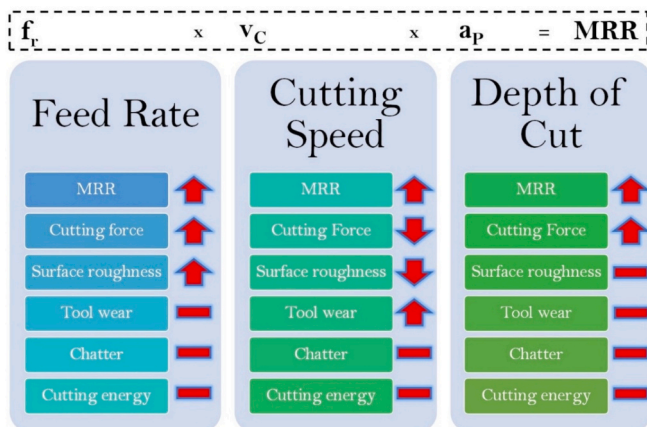


Fig. 3. Cutting parameters affecting MRR and their impacts on machinability variables.

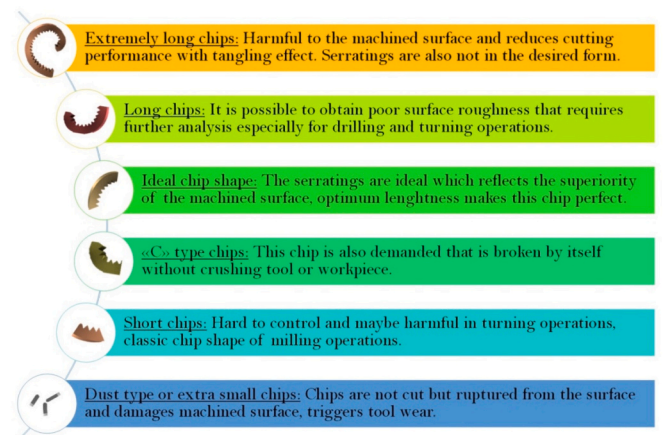


Fig. 4. General morphology of chips and possible effects on surface and machinability.

engineering products, a variety of chips are created during the different stages of the operations according to the treatment conditions, context of the material, and hardness value. This requires further investigation of the metals for recognition of the chip's morphology, color, and serrating structure.

Fortunately, tool manufacturers, aware of the huge problem caused by long ribbon-type chips, invest enormous amounts of money to design and produce ideal chips. They always have several chip breaker designs, that cover finishing medium and roughing cuts. Finding the right chip breaker, however, may not be an easy task due to the materials' diversity and their available properties, especially concerning steels.

4. Improvement of machinability of steels on different aspects

4.1. Improvement of machinability of steels with cutting parameters

According to Han et al. [111], there is an urgent need in the manufacturing processes, in general, to improve efficiency while reducing energy consumption. Regarding machining processes, Zerti et al. [112] also pointed out the importance of selecting optimal cutting conditions to achieve, at lower costs and in a short time, high levels in terms of accuracy. Additionally, Xu et al. [113] stated that optimizing cutting conditions can improve machining efficiency, resulting in economic benefits and energy saving. Thus, searching for optimal cutting conditions for a given workpiece and tool-type material plays an important role in increasing machining efficiency not only in terms of energy consumption or costs, but also in terms of integrity of the machined surface, tool wear, cutting forces, temperature, residual stresses, and even carbon emissions as proposed by Zhou et al. [114]. In this context, several works have been carried out focusing on determining optimal cutting parameters in machining processes of different types of steel.

In turning of AISI 52100 hardened steel (55 ± 2 HRC), Alok and Das [115] tested different cutting conditions in terms of cutting speed (v_c), feed rate (f), and depth of cut (a_p) and observed that v_c was the main influential factor for the cutting force, surface finishing, and the tool wear. According to the authors, cutting temperature increases with v_c , which leads to a reduction in the workpiece material's shear strength, thereby reducing machining forces. A combination of high v_c and low a_p values promoted better surface finishing. In terms of tool wear, the best results were achieved by lowering all the cutting parameters. Optimum parameters for minimizing machining forces, surface roughness and flank wear were $v_c = 195$ m/min, $f = 0.08$ mm/rev, and $a_p = 0.06$ mm.

Singh et al. [116] tested different cutting conditions on the dry turning of AISI 4340 and found that higher cutting speed contributed to reducing gradually the surface roughness. Higher depth of cut deteriorated surface finishing due to higher material removal rate (MRR) and cutting force. For the conditions tested, surface roughness increased sharply with feed rate, becoming almost constant for $f = 0.28$ mm/rev. Optimum cutting conditions were found as $v_c = 250$ m/min, $f = 0.20$ mm/rev, and $a_p = 0.5$ mm. Aslan [102] evaluated different cutting conditions in turning the AISI 5140 (32 ± 4 HRC). The findings showed that the tool wear was mainly affected by the variation in the cutting speed due to its effect in cutting temperatures. The depth of cut and the feed rate strongly affected machining force components. Optimized cutting parameters were found as $v_c = 500$ m/min, $f = 0.06$ mm/rev, and $a_p = 1.5$ mm.

Anandan et al. [117] evaluated the influence of the cutting parameters on turning the High-Speed Steel (HSS) M42. The authors found that both cutting temperature and the tool wear increased with feed rate and cutting speed. According to them, higher cutting speed improves surface finishing but deteriorates tool's sharpness due to accelerated wear because of higher cutting temperatures.

Barzegar et al. [118] analyzed the cutting temperature in face turning the AISI 1050 steel with different conditions of cutting speed (v_c) and feed rate (f). The main findings show that cutting temperature

significantly increases with both v_c and f . According to the authors, the strain rate and friction speed increase with v_c , thereby resulting in higher heat generation in all the deformation zones. Temperature levels measured in experimental trials varied in the range of 300°C to 430°C at the flank face, and in the range of 350°C to 450°C at the rake face.

Saez-de-Buruaga et al. [119] evaluated the tool/chip temperatures during orthogonal turning of four different ferrite-pearlite steel grades: 16MnCr5 (162 HB), 27MnCr5 (163 HB), C45 (185 HB), and C60 (188 HB). The findings show that increasing the cutting speed from 100 m/min to 200 m/min contributed to increasing the tool/chip contact temperature by 40 %. The increase in feed rate (0.1 mm to 0.2 mm) also increased the contact temperature, but less significant: 15 %. In general, temperature levels were in the range of 530°C to 1015°C , with differences between the steels lower than 130°C .

Rajaguru et al. [120] performed experiments of orthogonal dry turning of super duplex stains steel. The authors measured the feed and cutting forces, cutting temperature, chip thickness, shear angle, and residual stress. Different cutting conditions in terms of feed rate were tested. The results show that the cutting zone temperature increased with feed rate, varying in the range of 408 – 425°C . The feed and cutting forces, chip thickness, shear angle, and residual stresses also increased with the feed rate.

Wang et al. [121] studied the influence of cutting speed and feed rate on the cutting temperature, surface roughness, and cutting tool wear in turning the hardened bearing steel GCr15 (62–65 HRC) under low-temperature liquid CO_2 cooling-lubrication condition. The authors observed that the cutting temperature increased with both v_c and f . A similar tendency was observed for surface roughness, and optimum cutting conditions promoted up to 81.5 % reduction in Sa parameter. Tool wear increased with v_c and f , but the effect of v_c was more significant. Recommended parameters for machining quality and efficiency are $10\text{ }\mu\text{m/r} < f \leq 15\text{ }\mu\text{m/r}$ and $0.50\text{ m/s} < v_c \leq 1.00\text{ m/s}$, where r is the corner radius of the tool.

Li et al. [122] studied the machinability of stainless steel 316L fabricated by selective laser melting (SLM) in the turning process with tungsten carbide tools. Different cutting conditions were tested in terms of v_c and a_p . The main findings showed that, for $v_c = 80$ m/min, both cutting force and surface roughness (Ra parameter) increased with depth of cut. For the material fabricated by the SLM method, lower cutting forces and poor finishing were observed in comparison to the wrought material, except when using the highest depth of cut (0.6 mm). The authors observed that the thickness and hardness of the white layer were strongly influenced by thermal and mechanical loads, which are affected by the cutting parameters.

Wu et al. [123] analyzed the microstructure and mechanical properties of the surface and subsurface of the hardened steel P20 (55–56 HRC) after orthogonal cutting under cryogenic liquid nitrogen cooling method. Different cutting conditions in terms of cutting speed (50, 300, and 500 m/min) and tool wear (VB of 0 mm and 0.15 mm) were tested. The findings showed that both cutting speed and tool wear affected the surface integrity of the machined surface. Three different regions beneath the machined surface were identified: a strong plastic deformation white layer zone, a transition layer with moderate plastic deformation, and a region below the transition layer where elastic deformation takes place. According to the authors, no evidence of phase transformation nor recrystallization at white and transition layers was found for the conditions tested, thereby indicating that the formation mechanisms of these layers are related to severe dislocation and plastic deformation induced by tool wear and high cutting speeds.

In the milling process, Li et al. [124] evaluated the effect of the main cutting parameters on the energy efficiency and production cost when machining the AISI 1045 steel. The main findings showed that the specific energy consumption (SEC) always decreased when using higher feed rate and depth of cut values. Increasing v_c first decreases SEC, and then increases it after an inflection point. According to the authors, high cutting speed, feed rate, and depth of cut values reduce production time

and energy consumption. However, increasing the cutting speed also reduces tool life, thereby increasing the energy consumption for its changing. The authors present optimum conditions for minimum SEC and minimum production cost, mentioning the inherent trade-off between the material removal rate (MRR), energy efficiency, and production cost.

Li et al. [125] performed experimental trials in hard milling of AISI H13 steel (50 ± 1 HRC). Down milling and dry condition (no coolant) was adopted. Different cutting conditions in terms of cutting speed were evaluated. Considering the experimental results, the authors observed that the cutting temperature of the machined surface increased with cutting speed. Maximum temperature levels were recorded in the range of 446.8°C to 557.9°C . Work hardening and grain refinement with no phase transformation on the machined surface and beneath it were observed for the conditions tested by the authors.

Lopes et al. [126] evaluated the influence of different cutting parameters in milling a high-strength low alloy (HSLA) steel part produced by an additive manufacturing process. The main findings showed that using higher v_c in combination with low values of feed rate improved the surface finishing (lower values of R_a). According to the authors, higher values of v_c and f contribute to build-up edge (BUE) formation due to the low cutting temperatures and high levels of strain. Similar trends were observed for cutting force, which reduced when increasing v_c and increased with f . The authors pointed out that the microstructure characteristics of the material produced by additive manufacturing resulted in different tool wear rates in comparison to a near isotropic material, thus further investigation is needed. The findings also showed a reduction in chip thickness ratio by increasing the cutting speed, thereby indicating an improvement in the material's machinability. A 68 % reduction in R_a was reached by using the $v_c = 65$ m/min and $f_z = 0.0115$ mm/tooth.

Mokas et al. [127] evaluated the influence of different cutting conditions on the machinability of the annealed and hardened C22 (AISI 1020) steel in drilling. The findings showed that increasing both feed rate and depth of cut values reduced tool life. The authors observed that reducing the cutting speed increased the tool life when drilling the annealed material, while the opposite trend was observed for the hardened steel. The authors attributed this non-conventional result to changes in the microstructure of the hardened steel due to high-temperature levels at the contact zone for high values of v_c , as well as the reduction in chip thickness. Such results show that optimum cutting conditions also depend on material's microstructure characteristics, which corroborates the findings of Lopes et al. [126].

For grinding processes, Winter et al. [128] studied the influence of cutting depth (a_e), cutting speed (v_c), workpiece speed (v_w), and dressing speed (v_d) on several output parameters on grinding of the DIN 100Cr6 hardened steel. The findings showed that increasing a_e and v_w significantly increased specific cutting forces, cutting power, grinding energy, wheel wear, and surface roughness. Increasing v_c showed opposite trends, thereby decreasing all output parameters except for grinding energy. The authors pointed out the importance of considering a multi-objective optimization strategy to find optimum cutting conditions.

In a similar work, Deng et al. [129] evaluated the optimal cutting conditions for high efficiency and low carbon emissions for grinding AISI 1045 steel. The authors observed that higher feed speed and grinding depth values decreased the processing time, thereby reducing energy consumption and carbon emissions. However, high values of these parameters increased tool wear and negatively affected surface finishing.

Zhou et al. [130] observed that there is an improvement in the surface finishing by increasing the wheel speed (v_s) and lowering the feed speed (workspeed - v_w) during grinding of 12Cr2Ni4A gear steel. According to the authors, the number of active grits per unit of time increases with v_s , which reduces the residual height thereby forming a smoother surface. Increasing feed speed (or workpiece speed) increases

the distance of cutting-edge trajectories, which is not favorable for finishing. Similar results were also observed by Khare and Agarwal [131] during grinding of AISI 4340 hardened steel (65 HRC).

The above literature review shows that improving the machinability of steels with cutting parameters is possible, but generalization is challenging, especially considering the countless alloys, different tool materials, and cooling-lubrication conditions. Nevertheless, in general, an increase in cutting speed tends to improve surface finishing, although may increase tool wear due to higher cutting temperatures. Cutting forces usually increase with feed rate and depth of cut and reduce for higher cutting speeds. Increasing the feed rate and/or depth of cut values is positive in terms of material removal rate, which can result in economic benefits due to time processing reduction. However, the trade-off between the surface integrity of the workpiece, productivity, and economic aspects must be considered, and finding an optimal balance depends on research. Multi-objective optimization using statistical models and even artificial intelligent approaches can be interesting tools to complement experimental trials. Table 3 summarizes the references presented in this section.

4.2. Improvement of machinability of steels with cutting environment

The cutting environment can be understood as the cooling-lubrication conditions/strategy applied to the machining process, which includes cutting fluid type and its application technique, and all the input variables to control it such as cutting fluid concentration, flow rate, nozzle geometry/position, application pressure, temperature, and additives. A proper application of cutting fluids can allow the use of higher material removal rates, improve chip formation, tool life, surface finishing, and dimensional accuracy of the machined parts [132]. Sarikaya et al. [133] classified the cooling lubrication methods as shown in Fig. 5, and also specified the type of cutting fluid usually applied for each method and its usual flow rate as summarized in Table 4.

Dry machining is also a possible strategy, which is a desirable alternative in terms of economic and environmental aspects, and it can be used for machining steels. According to Liew et al. [134], dry machining is usually applied for turning hardened steels with highly resistant cutting tools such as polycrystalline cubic boron nitride (PCBN) and ceramics. Najiha et al. [135] highlighted the application of dry cutting in the machining of some specific titanium alloys, cast iron, and aluminum. However, this practice still presents challenges to manufacturers in general, since it often causes excessive temperature rises, thereby reducing the tool life and negatively affecting workpiece surface integrity Goindi and Sarkar [136].

In terms of conventional cooling (flood), de Paiva et al. [137] evaluated different oil-in-water concentrations in grinding the VP80 steel (45 HRC). The findings showed that increasing the oil concentration contributed to improving surface integrity of the ground part, especially for more severe cutting conditions. In a more recent work, de Paiva et al. [138] proposed controlling the cutting fluid temperature at low levels to improve the grindability of the hardened SAE 52100 steel (60 ± 2 HRC). Besides room temperature ($28 \pm 1^\circ\text{C}$), three low temperatures were tested: 5°C , 10°C , 15°C . The findings showed that using the cutting fluid at low temperatures avoided visual burns and reduced R_a roughness and the thermally affected zone by up to 31 % and 92 %, respectively. According to the authors, lowering the cutting fluid temperature improves its cooling and lubrication capacities, thereby promoting better temperature control at the contact zone during grinding.

The use of cooled cutting fluid via flood technique was also tested by Araújo et al. [139] in turning the AISI 1045 steel. The authors observed that this cooling-lubrication strategy reduced tool wear and improved workpiece finishing. The authors also pointed out that its cost is approximately 3 % of the cost of a cryogenic refrigeration system and 1.5 % of the cost of a high-pressure coolant (HPC).

Kramar et al. [140] tested the use of HPC in hard turning of the AISI 1045 steel with a 5.5 % concentration vegetable oil-based emulsion and

Table 3
Improvement of machinability of steels with cutting parameters.

Reference	Year	Process	Workpiece material	Tool	Input variables tested	Optimum cutting parameters	Output variables evaluated								
							Cutting force and moment	Energy consumption	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)	Chip morphology	Carbon emissions
[115]	2019	Turning	AISI 52100 hardened steel	Tungsten carbide PVD coated (HSN ²) CNMG 120408MBB	$v_c = 140\text{--}250$ m/min $f = 0.08\text{--}0.12$ mm/rev $a_p = 0.06\text{--}0.1$ mm	$v_c = 195$ m/min $f = 0.08$ mm/rev $a_p = 0.06$ mm	•			•		•			
[116]	2019	Turning	AISI 4340	yttria stabilized zirconia toughened alumina (Y-ZTA)	$v_c = 150\text{--}250$ m/min $f = 0.20\text{--}0.28$ mm/rev $a_p = 0.5\text{--}1.5$ mm	$v_c = 250$ m/min $f = 0.20$ mm/rev $a_p = 0.5$ mm	•			•				•	
[102]	2020	Turning	AISI 5140 (32 ± 4 HRC)	Tungsten carbide TCMT 16T304-P25	$v_c = 350$ and 500 m/min $f = 0.06$ and 0.1 mm/rev $a_p = 1.5$ and 2.0 mm	$v_c = 500$ m/min $f = 0.06$ mm/rev $a_p = 1.5$ mm	•					•		•	
[117]	2021	Turning	M42 HSS	Tungsten carbide CVD coated TNMG 160408MT	$v_c = 75\text{--}125$ m/min $f = 0.04\text{--}0.12$ mm/rev Environment (Dry, MQL, NMQL)	$v_c = 125$ m/min $f = 0.08$ mm/rev NMQL				•		•			
[118]	2021	Turning	AISI 1050	Tungsten carbide Uncoated	$v_c = 100\text{--}250$ m/min $f = 0.05$ and 0.10 mm/rev	–			•						
[119]	2018	Turning	16MnCr5 (162 HB), 27MnCr5 (163 HB), C45 (185 HB), C60 (188 HB)	Tungsten carbide (P25) Uncoated TPUN160308TTM	$v_c = 100\text{--}200$ m/min $f = 0.1\text{--}0.2$ mm	–	•		•						
[120]	2022	Turning	Super duplex stainless steel (SDSS)	Tungsten carbide PVD coated (AlTiN) Grooving insert A4G0505M05U04B	$f = 0.12\text{--}0.20$ mm/rev	–	•		•	•				•	
[121]	2023	Turning	GCr15 hardened steel (62–65 HRC)	PCD (55° rhombus, 7° clearance angle, 2 µm grain size, 3.97 mm thickness, 0° rake angle, 0.8 mm corner radius)	$v_c = 0.5\text{--}1.25$ m/s $f = 10\text{--}25$ µm/r	0.50 m/s < $v_c \leq 1.00$ m/s 10 µm/r < $f \leq 15$ µm/r			•	•		•			
[122]	2023	Turning	316L stainless steel	Tungsten carbide 0° rake angle and 7° relief angle	$v_c = 40\text{--}120$ m/min $a_p = 0.05\text{--}0.6$ mm	–	•			•				•	
[123]	2023	Turning	P20 Hardened steel (55–56 HRC)	PCBN (BK1011), 0° rake angle, 7° clearance angle, and negative chamfer (0.15 mm, –20°)	$v_c = 50, 300,$ and 500 m/min Tool wear (VB): 0 and 0.15 mm	–	•		•	•					

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Table 3 (continued)

Reference	Year	Process	Workpiece material	Tool	Input variables tested	Optimum cutting parameters	Output variables evaluated								
							Cutting force and moment	Energy consumption	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)	Chip morphology	Carbon emissions
[124]	2017	Milling	AISI 1045	GMA-125, Ø 125 mm (tool holder) Cemented carbide 6 inserts Lead angle: 12° Clearance angle: 15°	$v_c = 78.5, 157, \text{ and } 235.5 \text{ m/min}$ $f = 0.1, 0.2, \text{ and } 0.3 \text{ mm/tooth}$ $a_p = 1, 3, \text{ and } 5 \text{ mm}$ $a_e = 30, 60 \text{ and } 90 \text{ mm}$	–		•				•	•		
[125]	2018	Milling	AISI H13 hardened steel (50 ± 1 HRC)	Ti(C,N)- Al_2O_3 coated cemented carbide XOMX090308TR-M08 MP1500 (insert) R217.69-2020.0-09-3AN, Ø 20 mm (milling cutter)	$v_c = 200, 300, \text{ and } 400 \text{ m/min}$	–			•	•					
[126]	2020	Milling	HSLA - AM	Uncoated carbide end mill S902 6 mm diameter, 30° helix angle, 12° rake angle, 2 flute, with a chamfered cutting edge of 0.13 mm	$v_c = 30, 47.5, \text{ and } 65 \text{ m/min}$ $f = 0.0115, 0.023, \text{ and } 0.0345 \text{ mm/tooth}$	$v_c = 65 \text{ m/min}$ $f_z = 0.0115 \text{ mm/tooth}$ (lowest surface roughness)	•			•		•		•	
[127]	2018	Drilling	AISI 1020 (annealed and hardened)	high-speed steel twist drill of 10 mm and 20 mm diameter	$v_c = 11.15 \text{ and } 22.30 \text{ m/min}$ $f = 0.1 \text{ and } 0.2 \text{ mm/rev}$ $a_p = 10 \text{ and } 20 \text{ mm}$	–					•				
[128]	2014	Grinding	AISI 52100 hardened steel (62 HRC)	CBN grinding wheel Ø 40 mm (B 126 V T43 V 385)	$v_s = 40\text{--}60 \text{ m/s}$ $v_w = 0.40\text{--}0.70 \text{ m/s}$ $a_e = 0.80\text{--}3.50 \mu\text{m}$ $v_d = 23.22\text{--}43.32 \text{ m/s}$ Cooling-lubrication conditions (grinding oil, polymer dilution)	–	•	•		•		•			•
[129]	2016	Grinding	AISI 1045	Ø 200 mm, 25 mm thickness	$v_s = 21.99\text{--}31.14 \text{ m/s}$ $v_w = 1\text{--}4 \text{ m/min}$ $a_e = 10\text{--}40 \mu\text{m}$	$v_s = 21.99 \text{ m/s}$, $v_w = 1 \text{ m/min}$, $a_e = 10 \mu\text{m}$ (minimum surface roughness) $v_s = 21.99 \text{ m/s}$, $v_w = 4 \text{ m/min}$, $a_e = 40 \mu\text{m}$ (optimal carbon efficiency and shortest		•		•		•			•

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Table 3 (continued)

Reference	Year	Process	Workpiece material	Tool	Input variables tested	Optimum cutting parameters	Output variables evaluated						
							Cutting force and moment	Energy consumption	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)
[130]	2020	Grinding	12Cr2Ni4A gear steel	Microcrystalline corundum grinding wheel (30 % Cubitron 321) Ø 284 mm 80 (abrasive size), V (bond) F (hardness) 10 (structure number)	$v_s = 10\text{--}26\text{ m/s}$ $v_w = 1000\text{--}1800\text{ mm/min}$	—	—	—	—	—	—	—	—
[131]	2015	Grinding	AISI 4340 hardened steel (65 HRC)	—	$v_w = 5, 10, \text{ and } 15\text{ m/min}$ $a_e = 5, 10, \text{ and } 15\text{ }\mu\text{m}$	—	—	—	—	—	—	—	—

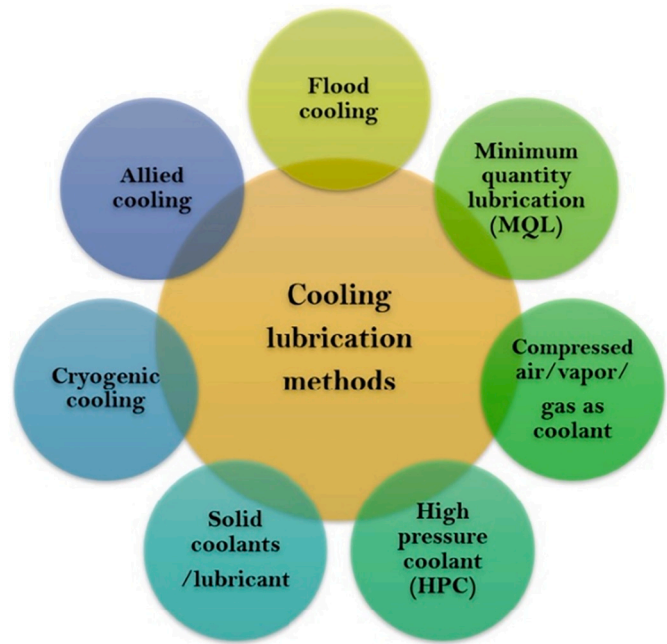


Fig. 5. Cooling lubrication methods [133].

Table 4
Cutting fluid/coolant and its flow rate for most common cooling/lubrication methods (adapted from [133]).

Cooling/lubrication method	Cutting fluid/coolant	Flow rate
Conventional cooling (flood)	Mineral, semisynthetic, and synthetic-based lubricants	0.5–10 L/min
MQL	Mineral, semisynthetic, and synthetic vegetable-based lubricants	10–500 mL/h
Cryogenic cooling	CO ₂ and LN ₂ (Liquid Nitrogen)	0.3–4 kg/min
HPC	Mineral, semisynthetic, and synthetic-based lubricants	10–100 L/min

reported a reduction in cutting forces compared to dry and conventional cooling. The chips' breakability was substantially improved for pressures higher than 70 MPa. Compared to conventional cooling, the authors observed a 400 % increase in tool life when turning with HPC and a 45 % and 25 % increase in maximum achievable cutting speed and feed rate values, respectively. Mia and Dhar [141] evaluated the application HPC in turning the AISI 1060 hardened steel with different tempering conditions (40 HRC, 48 HRC, and 56 HRC). The main findings showed that, compared to dry conditions, the use of HPC contributed to reducing surface roughness, cutting temperature, and tool wear. Similar results were found by Sredanović and Globočki Lakić [142] in turning the AISI 52100 hardened steel (62 HRC) with HPC. The authors pointed out that using HPC with carbide tools can be a good alternative in comparison to the application of CBN inserts in hard turning.

Khanna et al. [143] tested dry, flood, MQL technique, and cryogenic CO₂ environments in turning the 15-5 PHSS steel (345 HV). The findings showed that cryogenic turning reduced the cutting tool wear by 44 % and 68 % in comparison to flood and MQL techniques, respectively, as well as increased the cutting length as shown in Fig. 6(a). Power consumption was drastically reduced by turning under cryogenic cooling as shown in Fig. 6(b), which was attributed to a reduction in friction and the chip thickness, and the lower cutting tool wear as well.

Sterle et al. [144] studied the effects of LCO₂ cooling + MQL in drilling the 42CrMo4 steel (31 HRC). The main findings show that LCO₂ cooling improved chip evacuation but provided poor lubrication in

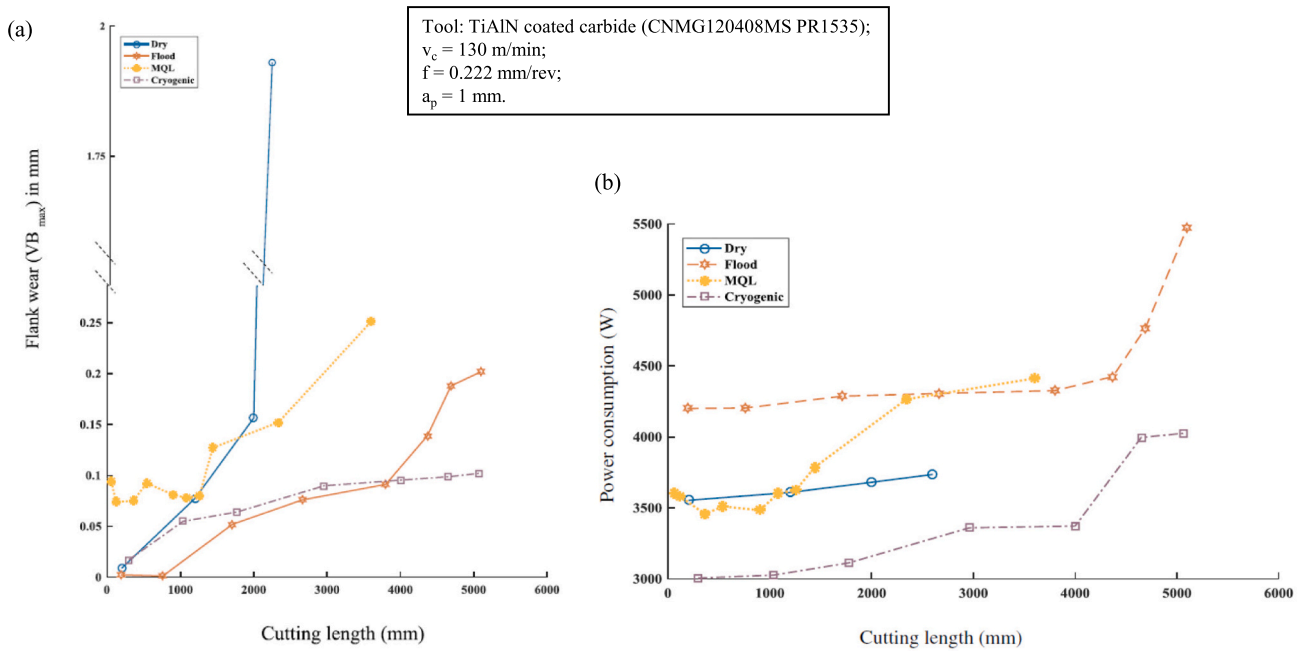


Fig. 6. Effect of cutting environment on flank wear (a) and power consumption (b) in turning the 15-5 PHSS steel. Adapted from [143].

comparison to the MQL technique, thus a proper balance between the two techniques must be found. Optimal cutting conditions to decrease LCO₂ consumption, increase productivity, and avoid clogging chips were $v_c = 70$ m/min, $q_{MQL} = 80$ mL/h, and $q_{LCO_2} = 115$ g/min.

Lai et al. [145] evaluated six different cooling methods in turning the 316L stainless steel: dry, flood, MQL technique, oil mixed with water droplets (OoW), supercritical carbon dioxide (Sc-CO₂), and Liquid nitrogen (LN). The authors observed that machinability improvement by the cooling method was strongly dependent on the type of tool's coating since each one presents different behavior in terms of wear resistance and wear characteristics. In general, the authors conclude that LN was the worse method in terms of tool wear. OoW was suitable for coatings with excellent resistance to crater wear and it is the better choice in terms of sustainability.

The use of the MQL technique with water has also shown promising results in grinding as observed by Sato et al. [146] in grinding the AISI 4340 hardened steel (54 ± 2 HRC). They found that a higher concentration of water in the cutting fluid contributed to improving the machinability in terms of roundness deviation and surface finishing. According to them, this is attributed to a reduction in clogging occurrence due to the lower viscosity of the cutting fluid with water, which is in good agreement to the findings of de Paiva et al. [147] in grinding the SAE 52100 hardened steel (60 ± 2 HRC).

Rajaguru and Arunachalam [148] investigated different cutting environments (dry, flood, and MQL) in turning the super duplex stainless steel (SDSS) 2507 with a water-miscible oil (1:20 dilution). The main findings show that the MQL technique contributed to reducing the tool wear, machining forces, the workpiece surface's roughness, residual stresses, and the chip reduction coefficient in comparison to dry and flood techniques. Additionally, the MQL technique contributed to improving the SDSS resistance to stress corrosion cracking.

Muaz and Choudhury [149] evaluated the use of solid lubricant-assisted cutting fluids applied with the MQL technique in milling the AISI 4340 steel. The solid lubricants tested were boric acid (B) and graphite (G) at a 10 wt% concentration. At optimum cutting parameters, the use of cutting fluid with solid lubricant contributed to reducing the resultant force. The best results were found for the MQL with boric acid.

The addition of nanoparticles in the cutting fluid to improve the MQL technique has also been extensively used in the machining processes of

steels, which is usually called nanofluid MQL (NMQL). Pal et al. [150] investigated graphene-enhanced cutting fluid in drilling the AISI 321 stainless steel (217 HB) and observed an improvement in the machinability in terms of thrust force, torque, surface finish, coefficient of friction, and tool wear. Similar results were found by Anandan et al. [151] in turning the M42 steel with graphene-enhanced oil NMQL. The authors observed that the NMQL technique contributed to reducing surface roughness, cutting temperature, and tool wear as shown in Fig. 7.

de Paiva et al. [152] evaluated the use of graphene-enhanced cutting fluid in grinding the SAE 52100 hardened steel (60 ± 2 HRC) and observed that low concentration of graphene (0.025 wt%) contributed to reducing surface finishing by 12 % in comparison to the traditional MQL technique (without nanoparticles). The ground surface's morphology was also improved by using graphene NMQL. In another work, de Paiva et al. [153] observed that the type of fluid has a significant effect on graphene NMQL by changing nanoparticle's tribological action mechanism. The authors also observed that the NMQL contributed to reducing the specific energy by up to 14 % in grinding the SAE 52100 hardened steel. Peña-Parás et al. [154] evaluated the addition of montmorillonite clay (MMT) NMQL in milling the AISI 1018 steel. The findings show that the NMQL can improve machinability in general by reducing spindle load, tool wear, and roughness of the machined surface due to a reduction in the contact area and the exfoliation process of MMT clay.

The above literature review shows that the most recent research regarding cutting environment is towards strategies that reduce cutting fluid usage/flow rate, even considering high-pressure cooling (HPC) methods. The preference for using the MQL technique is worth mentioning, especially considering the use of nanoparticles as additives that generally improve lubrication and heat dissipation capacities of the base fluids. The challenge is quite related to temperature and friction control at the contact zone, which can positively affect tool wear and the surface integrity of the machined part. Table 5 summarizes the references regarding the improvement of machinability with cutting environment presented in this review.

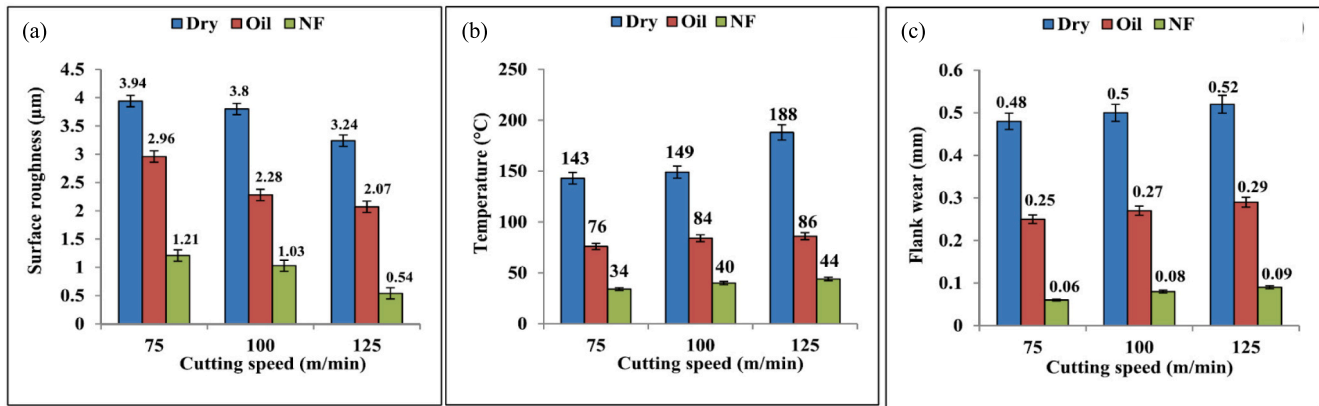


Fig. 7. Surface roughness (a), temperature (b), and flank wear (c) as a function of cutting speed value for different cutting environments (adapted from [151]). Workpiece material: M42 steel. Tool material: CVD coated carbide. Cutting parameters tested: $v_c = 75\text{--}125$ m/min, $f = 0.04\text{--}0.12$ mm/rev, $a_p = 0.4$ mm.

4.3. Improvement of machinability of steels with tool geometry and textured

According to Dogra et al. [155], tool geometry directly influences machining output variables such as surface integrity, machining forces, tool wear, heat generation, residual stresses, white/dark layer formation, and chip formation. Among the geometric tool parameters, the authors surveyed the rake angle, cutting edge, tool nose radius, rake face topology, and approaching angle. In general, the tool geometry can be inputted in almost all possible configurations depending on steel's properties, being in most cases a well-established knowledge.

The micromachining process remains the field with the least amount of knowledge regarding tool geometry influence in the machining process of steels. Aramcharoen and Mativenga [156] evaluated the effect of tool geometry and size in the micro-milling process of the H13 tool steel (45 HRC). The input parameter was the ratio between undeformed chip thickness to the cutting-edge radius at variable feed velocity (12–216 mm/min). According to the authors, the undeformed chip thickness is a key parameter, with the best results regarding surface finish found for the ratio between chip thickness and tool edge radius closer to 1, with the burr formation decreasing as this ratio increases.

As reviewed by Machado et al. [157], the most recently researched trend in tool geometry is the use of surface texturing to improve the overall machinability of the process. The authors further reported that steels had comprehended around 45 % of the researched papers on the topic since 2016; most of it is from hardened/high alloy steels (24 %), carbon/low alloy steel with 18 %, and only around 3 % of stainless steel.

Ma et al. [158] evaluated the performance of surface textured tools in dry turning the AISI 1045 steel, using the experimental results to create a finite element method model. The textures were composed of micro bumps in the rake face of the tool, with different edge distances (20–620 μm), width (70–300 μm), distance between individual patterns (20–200 μm), and width/height ratios (2–16). The authors reported that the textures reduced the cutting force by 10 % and the thrust force by 20 %. The lowest cutting forces were observed for the widths of 20 and 400 μm, showing the complexity of the interactions of this tribosystem. The best distance from the cutting edge was found between 320 and 420 μm. Duan et al. [159] studied the derivative cutting at a single and parallel to the cutting-edge linear groove in the dry turning of the AISI 1045 steel. The authors reported that the derivative cutting increased the chip filling of the texture, increasing friction and cutting forces. The derivative cutting damaged the cutting edges, and larger negative rake angles can attenuate this phenomenon.

Sivaiah [160] studied the performance of dimples, grooves, and hybrid textures composed of booth on WC/Co AlTiN coated tools, in the machining process of the AISI 304 stainless steel under the MQL environment. The results showed that the hybrid texture improved tool

wear, roughness, and machining temperature compared to the individual patterns. Vasumathy and Meena [161] investigated the use of surface textured WC/Co tools in the machining of AISI 316 stainless steel. The textures were laser-manufactured orientations parallel, perpendicular, inclined with acute, and inclined with obtuse angles related to the cutting edge. The results indicated that texture directly influences the tool/workpiece friction, with the best machinability regarding machining forces, tool wear, and chip formation achieved by the parallel texture.

Zhang et al. [162] studied nano and micro-scale textures on WC/Co-AlTiN tools in the turning process of AISI 316 stainless steel. The textures with a considerably smaller scale compared to the current literature were obtained using a femtosecond Nd: YAG (Neodymium-doped Yttrium Aluminum Garnet) laser before the coating process. The surface of the tool was laser-treated (nano-texturing) to increase coating adhesion. The authors reported that both the micro and micro/nano textures improved tool life in comparison to standard tools, with the improved adhesiveness of nano-textured tools providing better results than the micro-textured tools regarding tool wear.

Rajbongshi et al. [163] studied the use of conventional and textured WC-Co tools in the dry machining of the AISI D2 (45 HRC) steel. The results indicated that textured tools outperformed conventional ones regarding surface integrity, flank wear, and machining temperatures. In a similar study, Duan et al. [164] evaluated the surface textures in the dry machining of the AISI H13 tool steel (48–52 HRC). The results pointed out that, despite the derivative cutting (Fig. 8) leading to cutting force and temperature increase, it is beneficial for the chip formation as it increases the plowing at the interface between the chip and the tool, leading to a higher chip segmentation and thus improving its breakage.

Law et al. [165] studied the machinability of hardened bearing steel (56 HRC) in the dry finish turning process using laser textured (dimples and grooves) PcbN tools. The results pointed out that tool life, surface quality, and machining forces did not increase significantly with textured tools. The authors further state that using textures near the tool edge even increased the tool wear. Xing et al. [166] evaluated the use of nanoscale and microscale textures manufactured at Al₂O₃/TiC coated carbide tool with and without solid lubricant (MoS₂) burnished at the surface textures in the dry turning of a hardened AISI 1045 steel (40–50 HRC). The authors reported that using textures improved the machinability regarding machining force and temperature, as well as tool friction and wear, with the results further improved using the nano-scale texture. The use of self-lubricating tools increased even further the machinability of the system.

Summarizing the content of this section, it was presented a discussion about the influence of tool geometry on the machining process of steels. It covered various studies that have explored the impact of different tool parameters on machining output variables, including

Table 5
Improvement of machinability of steels with cutting environment.

Reference	Year	Process	Workpiece material	Tool	Cutting fluid/ application technique	Cutting environments tested	Output variables evaluated								
							Cutting force and moment	Energy consumption	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)	Chip morphology	Carbon emissions
[137]	2017	Grinding	VP80 (45 HRC)	Al ₂ O ₃ – AA46K6V (Ø 280 mm)	Water miscible – synthetic oil Flood (9 L/min)	Cutting fluid concentration: 3 and 8 %				•					
[138]	2021	Grinding	AISI 52100 (60 ± 2 HRC)	Al ₂ O ₃ – 38A46K6V (Ø 297 mm)	Water miscible – semisynthetic vegetable-based oil (5 % concentration)	Cutting fluid temperature: 5 °C, 10 °C, 15 °C, and 28 ± 1 °C				•					
[139]	2019	Turning	AISI 1045	Coated carbide KNUX 160410R11	Water miscible – synthetic oil (ratio of 20:1) Flood	Cutting fluid temperature: Refrigerated (10–12 °C) and non-refrigerated				•					
[140]	2010	Turning	AISI 1045 (58 HRC)	Al ₂ O ₃ -coated carbide SNMA 120408 KR 432	5.5 % vegetable oil-based emulsion	Dry, Flood, and HPC at different pressures: 10–200 MPa	•				•		•	•	
[141]	2017	Turning	AISI 1060 (40, 48, and 56 HRC)	Coated carbide SNMG 120408	–	Dry, HPC (80 bar, 6 L/min)			•	•		•			
[142]	2017	Turning	AISI 52100 (62 HRC)	TiSiN–TiAlN PVD coated carbide CNMG 1204 08 MF5 TH1000	Water miscible – vegetable-based oil (3 % concentration)	HPC (50 MPa, 2 L/min)				•	•		•		
[143]	2020	Turning	15-5 PHSS	PVD Mega-Nanocoat carbide (TiAlN) CNMG120408MS PR1535	5 % mineral oil emulsion (Flood) Neat synthetic oil (MQL) LCO ₂ (cryogenic)	Dry, Flood (3.75 L/min), MQL (14 mL/h, 6 bar), and Cryogenic	•				•	•			
[144]	2021	Drilling	42CrMo4 steel (31 HRC)	Solid carbide drills AlCrTiN coating and internal cooling channels D = 4 mm, lip clearance angle = 140°	Neat oil (MQL) LCO ₂ (cryogenic)	LCO ₂ : 0, 125, 250 g/min MQL: 0, 50, 100 mL/h							•	•	
[145]	2020	Turning	316L stainless steel	Carbide inserts Different AlTiN-based coatings	Oil-based cutting fluid (Flood) Micro biodegradable lubricant (MQL) Micro biodegradable lubricant and water (OoW) Supercritical carbon dioxide (Sc-CO ₂)	Dry Flood MQL (60 mL/h, 0.6 MPa) OoW: Oil (60 mL/h), water (150 mL/h) and compressed air (0.6 MPa) Sc-CO ₂ (5.5 L/h, 8.5–9 MPa, –68 °C) LN (11.75 L/h, –174 °C)						•			

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Table 5 (continued)

Reference	Year	Process	Workpiece material	Tool	Cutting fluid/ application technique	Cutting environments tested	Output variables evaluated								
							Cutting force and moment	Energy consumption	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)	Chip morphology	Carbon emissions
[146]	2020	Grinding	AISI 4340 (54 ± 2 HRC)	Vitrified CBN wheel SBN 151 Q12 VR2 Ø 350 mm	Liquid nitrogen (LN) 3 % emulsion in water (Flood) Synthetic oil (MQL)	Flood MQL (100 mL/h, 6.5 MPa) with different cutting fluid concentrations (0 %, 50 %, 75 %, and 83.34 % water in oil)				•					
[147]	2021	Grinding	AISI 52100 (60 ± 2 HRC)	Al ₂ O ₃ – 38A60K6V (Ø 250 mm)	Semisynthetic vegetable-based and Synthetic 1:20 dilution for Flood Neat oil for MQL	Flood (9 L/min) MQL (150 mL/h, 0.3 MPa)				•					
[148]	2020	Turning	SDSS 2507	Multilayer coating (MT-TiCN/Al ₂ O ₃) tungsten carbide (TNMG 160408)	Water miscible oil (1:20)	Dry, Flood, and MQL (150 mL/h, 0.6 MPa)	•			•		•		•	
[149]	2019	Milling	AISI 4340	Tungsten carbide CVD coated (TiCN/Al ₂ O ₃ /TiN) Ø 16 mm, 2 teeth	Emulsion (ratio of 1:20) Emulsion + 10 wt% Boric acid Emulsion + 10 wt% Graphite Emulsion + 10 wt% Boric acid + 10 wt% Graphite	MQL, MQL + solid lubricants (150 mL/h, 5 MPa)	•			•					
[150]	2020	Drilling	AISI 321 (217 HB)	M35 HSS (Ø 8 mm) 37° helix angle, 118° point angle	Sunflower oil (MQL) Sunflower oil + graphene (NMQL)	MQL and NMQL at different concentrations (0.5–1.5 wt%) (120 mL/h, 6 bar)	•			•		•	•		
[151]	2021	Turning	M42 HSS	CVD coated carbide TNMG 160408MT	Canola oil (MQL) Ethylene glycol + graphene (NMQL)	Dry, MQL and NMQL (graphene) (10 mL/min, 1.5 bar)			•	•		•			
[152]	2020	Grinding	AISI 52100 (60 ± 2 HRC)	Al ₂ O ₃ – 38A60K6V Ø 250 mm	Semi-synthetic vegetable-based oil (MQL) Semi-synthetic vegetable-based oil + graphene (NMQL)	MQL and NMQL at different concentrations (0.025, 0.050, and 0.075 wt%) (150 mL/h, 0.3 MPa)		•		•					
[153]	2023	Grinding	AISI 52100 (60 ± 2 HRC)	Al ₂ O ₃ – 38A46K6V Ø 250 mm	Semisynthetic vegetable-based and Synthetic 1:20 dilution	Flood (9 L/min) MQL and NMQL (150 mL/h, 0.3 MPa)		•		•					

(continued on next page)

Table 5 (continued)

Reference	Year	Process	Workpiece material	Tool	Cutting fluid/ application technique	Cutting environments tested	Output variables evaluated								
							Cutting force and moment	Energy consumption	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)	Chip morphology	Carbon emissions
[154]	2019	Milling	AISI 1018	Cemented carbides coated with TiAlN FK080-45-06-1 6E face shell mill (1 insert)	(Flood) Neat oil (MQL) Neat oil + graphene (NMQL) water miscible mineral oil, 14 % oil concentration (MQL) water miscible mineral oil, 14 % oil concentration + MMT (NMQL)	MQL and NMQL (0.1, 0.2, 0.3, and 0.4 wt%)	•	•	•	•	•	•	•	•	•

surface integrity, machining forces, tool wear, heat generation, residual stresses, and chip formation. The research delves into both conventional and micromachining processes, with a focus on the use of surface texturing to enhance machinability. The studies demonstrate that the careful application of surface textures, such as micro bumps, grooves, and dimples, can lead to significant improvements in cutting forces, tool wear, and surface finish. However, the effectiveness of specific textures varies depending on factors like tool material, cutting speed, and steel hardness. While some studies show promising results, others indicate that certain texturing approaches may not always deliver the expected benefits. Nonetheless, research in this area continues to advance, aiming to optimize tool geometry for improved steel machining outcomes. A summary of the presented papers is given in Table 6.

4.4. Improvement of machinability of steels with tool material including coating

Although cemented carbide tools are still the most commonly cutting tool materials employed in machining steels, the use of other materials such as cubic boron nitride (CBN) and ceramics has been increasing, especially considering the machining of difficult-to-cut steel alloys. Sobiya et al. [167] compared CBN and mixed ceramics in the dry machining process of the AISI 440B stainless steel. The results showed that the change in tool material resulted in a change in the main wear mode, with abrasion for the ceramic and adhesion for the CBN tool, with a better surface finish for the ceramic tool, despite the accelerated wear when compared to the CBN.

Sales et al. [168] compared TiN-coated WC-Co, mixed ceramic, and pCBN tool materials in the dry machining process of the W320 hardened steel. The results pointed out that at lower cutting speeds the cemented carbide presented a better performance, with the PCBN tools outperforming the other tool materials at cutting speeds higher than 200 m/min, with the mixed ceramic underperforming in all evaluated conditions, mainly due to its low toughness. Chen et al. [169] evaluated the use of TiN/Al₂O₃/Ti(C,N) coating on cermet tools in the machining process of a quenched and tempered steel (61–62 HRC). The results pointed out the uncoated cermet with greater tool life at the lower depths of cuts; however, the cutting forces increased enough to make the coated tool outperform its uncoated counterpart at more significant depths. The results further showed that coated tools lead to a slightly improved surface finish and chip formation.

The primary focus in augmenting the machining capabilities of cemented carbide tools is the development of hard coatings. A study conducted by Che Haron et al. [170] compared the performance of coated carbide tools (with TiCN, Al₂O₃, and TiN) with uncoated ones during the turning operation of 95MnCrW steel (23 HRC). The coated tools demonstrated superior wear resistance and longer tool lives than the uncoated ones. Another study by Lahres et al. [171] investigated the performance of various coatings (including Ti(C,N), TiAlN+MoS₂, (Ti, Al)N (single layer), TiAlOx) in the milling operation of 22Mn6 steel. The authors' tests of tool life resulted in a ranking of coatings and cooling atmospheres, arranged from the most effective to the least in the following order: MQL + TiAlN+MoS₂, Dry + TiAlN+MoS₂, MQL + Ti(C, N), Dry + Ti(C,N), and Flood + Ti(C,N).

Endrino et al. [172] carried out a comparative study of AlTiN, AlCrNbN, and AlCrN PVD (Physical Vapor Deposition) nano and fine crystalline coatings applied to cemented carbide tools during the milling of AISI 316 stainless steel. They found that the nanocrystalline AlTiN coating surpassed the others in terms of tool life. Fox-Rabinovich et al. [173] assessed the performance of single-layer and nanocrystalline TiAl₅₅CrSiYN, as well as nanocrystalline and multi-layer TiAlCrSiYN/TiAlCrN coatings on carbide tools in the ball-nose milling of AISI H13 hardened steel. Their findings highlighted that the multi-layer TiAlCrSiYN/TiAlCrN coating outperformed the others in terms of tool life, cutting forces and temperature, and chip formation.

Kumar et al. [174] compared two methods for depositing TiAlN

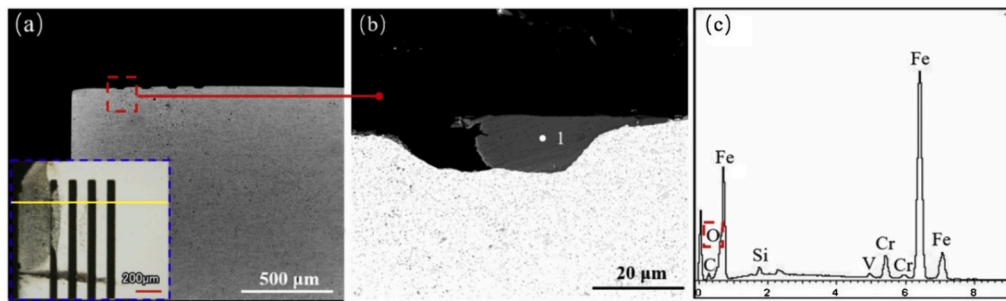


Fig. 8. (a) Cross-section of the textured tool; (b) region with derivative cutting; (c) chemical composition of the adhered material indicated by point 1 [164]. Workpiece material: AISI H13 hardened steel. Tool material: solid cement carbide, YS 8. Cutting parameters tested: $v_c = 95\text{--}237$ m/min, $f = 0.198$ mm/rev, $a_p = 0.5$ mm.

coatings (high power impulse magnetron sputtering - HiPIMS and direct current magnetron sputtering - DCMS) on cemented carbide tools during high-speed (600 m/min) dry turning of C20 and C80 carbon steels. They reported superior tool wear performance with the HiPIMS coatings due to denser and harder deposits. He et al. [61] analyzed the interface between the tool and two kinds of coated carbide tools (CVD - TiCN + Al_2O_3 and PVD - AlTiN) in the dry turning of AISI 304 stainless steel (Fig. 9). They found that the CVD coatings provided superior machinability owing to better cohesion between the coating and substrate. In particular, the CVD 1 coating was more effective than all the others. They further noted that the thickness of the AlTiN PVD coating did not significantly affect the tool's wear behavior.

Aramcharoen et al. [175] assessed the effectiveness of various coatings, including TiN, CrN, TiAlN, TiCN, and CrTiAlN, on tools used in the micro-slot milling of hardened H13 steel (45 HRC). They found that every coating improved the wear resistance of the tools when compared to those without any coating. The TiN coating showcased the best results in terms of flank wear resistance. However, it was found that only the TiAlN coating led to an increased burr formation than the tools without any coating. K. Zhang et al. [176] analyzed the efficiency of laser-textured substrates of WC-CO tools in enhancing the TiAlN coating's adhesion (as depicted in Fig. 10) during the dry milling process of AISI 316L stainless steel. The research team found that the implementation of nanotextures led to an improvement in the coating's adhesion. This enhancement subsequently resulted in a decrease in tool wear, cutting temperature, and cutting forces while enhancing surface roughness.

The literature review presented in this section showed that the selection of the appropriate tool material is a crucial factor in achieving successful steel machining. Striking the right balance between hardness and toughness is essential to enhance tool wear resistance and prevent chipping, ultimately leading to extended tool life. Tool hardness holds greater significance in continuous cutting operations like turning and drilling, as well as in machining hardened steels during finishing conditions. On the other hand, toughness plays a more critical role in interrupted cutting processes such as milling and roughing cuts. Tool coatings offer a favorable solution for machining steels using cemented carbide and high-speed steels (HSS), as they significantly improve tool lifespan and enhance the surface quality of the workpiece. These coatings can be applied through Chemical Vapor Deposition (CVD) or Physical Vapor Deposition (PVD) methods. CVD coatings, performed at higher temperatures (around 1000 °C), are suitable for continuous cutting applications, because they may reduce coating adhesion and result in the formation of the "eta phase" ($\text{Co}_3\text{W}_3\text{C}$), thus impacting tool toughness. On the other hand, PVD coatings are applied at lower temperatures (around 500 °C) being used for HSS, and in cemented carbides the "eta phase" is not formed, maintaining higher levels of tool toughness. This makes PVD-coated tools more suitable for interrupted cutting processes. A synopsis of these studies and their findings has been compiled in Table 7.

4.5. Improvement of machinability of steels with vibration, thermally and hybrid assisted machining

Zhou et al. [177] described vibration-assisted machining (VAM) as a process that employs high-frequency vibration to modify the tool's contact time during chip formation, thereby enhancing the chip formation process. Yang et al. [178] revealed that VAM is primarily employed in linear (1D) and circular/elliptical (2D) tool trajectories, and the vibration is predominantly introduced through sonotrodes (consisting of a horn & booster) or piezoelectric resonators.

Sharma and Pandey [179] studied how vibration parameters impact the ultrasonic assisted turning (UAT) process when working with hardened 4340 steel. Their findings indicated that while the feed rate and cutting speed substantially contributed to the reduction of residual stresses (measured by X-ray diffraction - XRD), the vibration amplitude emerged as the parameter with the strongest direct correlation with the decrease in residual stress. In a similar vein, Willert et al. [180] evaluated the impact of UAT with the elliptical vibration mode on a comparable material, AISI 4140 hardened steel, in both quenched-tempered (47 HRC) and annealed (25 HRC) states. Their observations suggested that implementing UAT resulted in decreased plastic deformation and surface roughness at the machined interface, regardless of the type of tool material used.

Zhang et al. [181] assessed the influence of 1D and 2D VAM on machining forces, temperatures, and tool wear during the diamond turning process of the AISI 1045 (ASSAB 760) carbon steel. They found that due to the low thermal input of the process, the difference in temperature between conventional and VAM methods was minuscule, with analogous results noted for machining energy consumption. However, they reported that both 1D and 2D VAM lessened tool wear, especially when air flow was directed towards the cutting region.

Arefin et al. [182] studied the effects of 1D VAM on chip formation during the turning process of the SKD11 tool steel (at 40, 50, and 60 HRC hardness levels). They conducted both experimental studies and finite element analysis for this purpose. Their results, which are depicted in Fig. 11, indicated that utilizing VAM led to a reduction in chip thickness, and an increase in the chip compression ratio, curling, and breakability.

Another machining strategy is the thermally assisted process, which aims to reduce the material's strength thereby facilitating chip formation. Baek et al. [183] described thermally assisted machining as a method that typically uses lasers (LAM - laser-assisted machining) or another heat source close to the cutting zone. This process softens the material set to be machined, subsequently reducing cutting forces. However, they also noted that working with a softer, hotter material could potentially lower the quality of the surface finish and amplify tool wear. In the same study, Baek et al. [183] examined the machinability of annealed AISI 1045 steel under various material removal rates during conventional, induction, and laser-induction-assisted machining (milling operation). Their findings indicated a decrease in cutting forces up to 83 % for induction-assisted and 65 % for laser-assisted machining in

Table 6

Improvement of machinability of steels with tool geometry and texture.

Reference	Year	Process	Workpiece material	Tool	Input variables tested in terms of geometry/texture	Output variables evaluated						
						Cutting force and moment	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)	Chip morphology
[156]	2009	Micromilling	H13 tool steel (45 HRC)	Tungsten carbide Two-flute microflat end mills ($\varnothing$ 900 μ m)	Ratio of undeformed chip thickness to tool edge radius r_e : 0.2–2.0			•				•
[158]	2015	Turning	AISI 1045 steel	WC/Co cemented carbide	Microbumps on the rake surface with different parameter values	•						
[159]	2017	Turning	AISI 1045 steel (19–31 HRC)	WC/Co cemented carbide	Textured tool (single linear groove on the rake face) and untextured tool	•				•		•
[160]	2019	Turning	AISI 304 stainless steel	AlTiN PVD coated tungsten carbide CNMA432 KC 910	Circular pit holes texture Circular pit holes and linear grooves hybrid texture		•	•		•		
[161]	2017	Turning	AISI 316 stainless steel	Uncoated WC-Co cemented carbide SPUN 120308	Untextured and four different micro-scale rectangular grooves textures (PA, PE, IN1, and IN2)	•				•		•
[162]	2015	Turning	AISI 316 stainless steel (198 HRB)	WC/Co-AlTiN	Conventional TiAlN coated tool without texture (CCT) Micro-scale textured TiAlN coated tool (MCT) Micro/nano-scale textured TiAlN coated tool (MNCT)					•		
[163]	2018	Turning	AISI D2 (45 HRC)	Coated carbide SNMG 12 04 08-KM3215	Textured tool (micro-EDM linear micro grooves on the flank face) Untextured tool		•	•		•		
[164]	2019	Turning	AISI H13 (48–52 HRC)	Solid cemented carbide (type YS 8, $16 \times 16 \times 5$ mm ³ three dimensions)	Textured tool (femtosecond laser, microgrooves of 1 mm long separated by 100 μ m, rake face) Untextured tool	•						•
[165]	2020	Turning	AISI 52100 (56 $\pm$ 3 HRC)	PcBN (CNGA 120408S-01020-L1-B CBN010)	Textured tools (grooves and dimples with different configurations, on rake face, flank face, secondary flank face, and chamfer)	•		•				
[166]	2014	Turning	AISI 1045 (40–50 HRC)	Hot-pressed Al ₂ O ₃ /TiC ceramic	Untextured tool Textured self-lubricated tools with different	•	•			•		

(continued on next page)

Table 6 (continued)

Reference	Year	Process	Workpiece material	Tool	Input variables tested in terms of geometry/ texture	Output variables evaluated					
						Cutting force and moment	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)
				12 × 12 × 7.94 mm ³ , 0.1 mm at 5° edge chamfer, 0.1 mm nose radius	configurations (perpendicular and parallel linear grooves, and wavy grooves) Untextured tool						

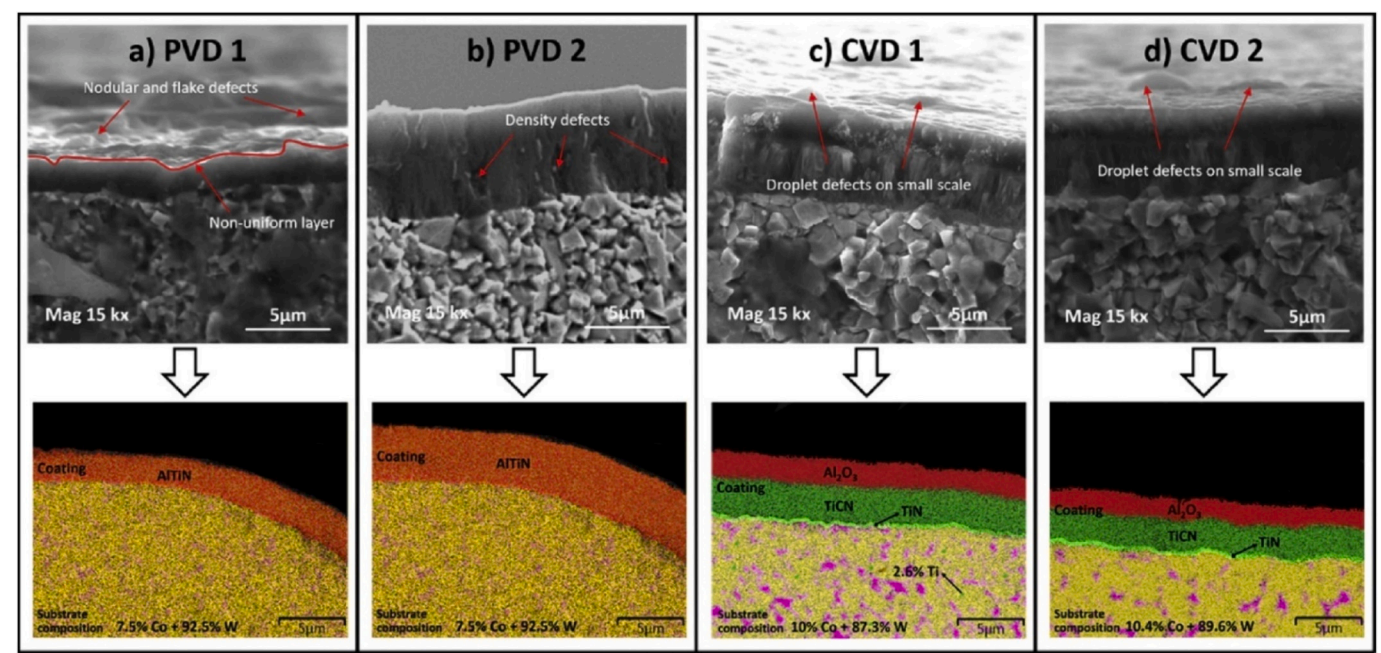


Fig. 9. Coatings evaluated by [61]. Tool material: cemented carbide. Tool coatings tested: PVD AlTiN and CVD TiN-MT-TiCN-Al₂O₃.

comparison with conventional machining. They further demonstrated in Fig. 12 that both thermally assisted methods resulted in reduced tool wear and improved surface finish quality.

Ahn et al. [184] investigated the effect of LAM on the specific cutting energy at different material removal rates (MMR) during the milling process of annealed AISI 1045 steel. Their research concluded that the implementation of LAM led to a reduction in cutting forces, with decreases in the specific cutting energy by up to 60 %. Ding and Shin [185] evaluated the surface integrity of hardened AISI 4340 steel samples (44–50 HRC) that were machined using both conventional and LAM techniques. They reported that the use of LAM lessened cutting forces and specific cutting energy by up to 20 %. Furthermore, they found that LAM led to larger compressive stresses in regions up to 40–50 µm from the surface.

Panjehpour et al. [186] researched the implementation of LAM in the hard-turning process of AISI 52100 steel (54 HRC). The variables under consideration in their experiment were cutting speed, feed rate, laser power, pulse frequency, and energy. Among their key findings was that LAM reduced the specific cutting energy and surface roughness by up to 25 % and 18 %, respectively. Additionally, they found that LAM led to more uniform plastic deformation of the chip when compared to conventional machining.

Bermingham et al. [187] studied the impacts of Laser Assisted Machining (LAM) on the cutting forces and tool wear during the milling process of 17-4PH precipitation-hardened stainless steel. They found

that LAM significantly decreased wear rates and cutting forces by as much as 50 % and 33 %, respectively. The authors reasoned that this reduction in cutting forces was a result of the workpiece’s thermal softening at the cutting interface, while the decreased tool wear was due to the reduced abrasive and adhesive action resulting from the softened machined material. In a similar study, Cao et al. [188] explored the effects of LAM in the milling process of 13-8 precipitation-hardened stainless steel. They concluded that the LAM process led to a 20.1 % reduction in cutting forces and a 34.4 % improvement in surface roughness.

A machining procedure is termed “hybrid” when a combination of techniques delivers an output effect greater than the sum of the individual techniques’ outputs, i.e., the techniques act synergistically [189]. Farahnakian and Razfar [190] researched a hybrid machining process that employed both ultrasonic and plasma technologies to assist in the turning process of the hardened AISI 4140 steel. Their analysis evaluated parameters such as cutting speed, depth of cut, feed rate, tool material, and generic plasma and ultrasonic factors, and the results were compared against conventional, plasma, and vibration-assisted machining. They found that the hybrid machining process resulted in lower cutting forces (5–60 %) and a superior surface finish (5–20 %) when compared to other machining conditions.

Liang et al. [191] studied the hybrid process that integrated ultrasonic cavitation and vibration in the micro-drilling process of AISI 304 steel. They found that, in general, the hybrid drilling process

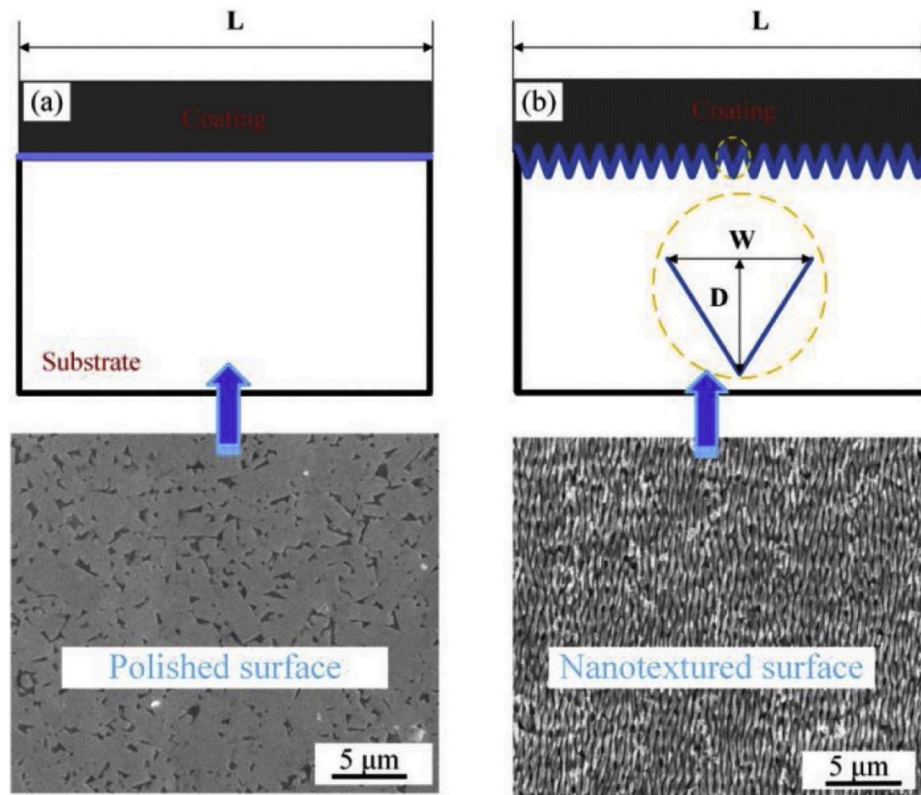


Fig. 10. Nanotextures manufactured by [176] aimed to increase coating adhesion compared to a polished surface. Tool material: TiAlN PVD coated WC/Co.

outperformed all other methods. In terms of tool wear, chip morphology, cutting forces, and hole morphology (as depicted in Fig. 13), Ultrasonic Cavitation Drilling (UCD) came in second, followed by Ultrasonic Vibration Drilling (UVD), and finally, Conventional Drilling (CD).

The beforementioned research discussed various techniques employed to improve the machining of steels. Vibration-assisted machining (VAM) utilizes high-frequency vibrations to modify the tool's contact time during chip formation, enhancing chip formation. Ultrasonic assisted turning (UAT) and thermally assisted machining are other strategies explored. UAT, employing vibration, reduces residual stresses, while thermally assisted machining uses lasers or heat sources to soften materials and reduce cutting forces. Hybrid machining, combining ultrasonic cavitation and vibration, has also shown improved results in terms of cutting forces and surface finish in micro-drilling processes. These techniques offer the potential for optimizing the machining of steels, depending on specific requirements and materials used. A summary of these studies is given in Table 8.

5. Challenges and future trends

A vast range of steel types complicates generalizations about machinability. From easily cut to hard-to-cut varieties, steel spans a wide spectrum. Comprehensive studies focus on specific grades with unique characteristics. Mechanical properties like ultimate tensile strength, hardness, work hardening, and ductility can predict machining behavior. Balancing hardness and ductility is crucial for optimal machining performance, as high hardness often means low ductility, and vice versa.

Inconsistent machining with materials from different batches is caused by slight variations in chemical elements, disrupting productivity by necessitating adjustments to cutting conditions. Achieving desired surface quality for steel components is complex due to diverse properties, requiring a deep understanding of their impact on surface integrity. Using AI tools alongside experimental testing is recommended to

comprehensively understand the effects of different steel properties on surface quality and develop effective machining strategies.

Despite advancements in the theoretical design of machining processes for steels, several challenges remain. One key challenge is the optimization of cutting parameters, which requires extensive knowledge and experience tailored to the specific steel being machined. Furthermore, variations in cutting tools, including tool geometry, material, and coating strategies, need to be critically examined. While numerous cutting tool manufacturers exist, they may offer seemingly similar products; however, their performance may differ significantly.

The optimization of Material Removal Rate (MRR) is a crucial consideration in the cutting of steels. Determining the ideal MRR holds significant importance as these materials play a vital role in the production of everyday products. In this regard, it becomes essential to thoroughly discuss and analyze different types of steels to make informed decisions based on their specific applications. Some industrial sectors require better surface quality, other sectors want corrosion resistance, others mechanical strength and so on; this will strongly affect the selection of the cutting parameters and thus the MRR.

Maybe the biggest problem in the continuous machining of steel is “chip control”. The challenges involved in chip classification and characterization include the need to address the advantages and difficulties associated with different types of chips. Long chips have the potential to tangle the cutting tool holder and workpiece material, disrupt the material removal process, increase friction and tool wear, and negatively impact surface quality. Further investigation of metal types is required to understand chip morphology, color, and serrating structure, which is a challenge, considering the wide range of engineering products and steels available.

It is feasible to enhance the machinability of steels through the selection of optimal cutting parameters. However, generalizing the findings becomes a complex task due to the vast array of alloys available, each possessing unique mechanical properties. Furthermore, the inclusion of various cutting tool materials and cooling strategies further adds

Table 7

Improvement of machinability of steels with tool material including coating.

Reference	Year	Process	Workpiece material	Tool material/coated tested	Cutting tool designation/geometry	Output variables evaluated						
						Cutting force and moment	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)	Chip morphology
[167]	2015	Turning	AISI 440B stainless steel (40–44 HRC)	Mixed ceramic (CC 650): Al ₂ O ₃ (70 %) + TiN (22.5 %) + TiC (7 %) average grain size of TiC is 4 µm	SNGN 090308S-01020			•		•		
[168]	2009	Turning	W320 hardened steel	PcBN (CBN 100): cBN (50 %) + TiC (40 %) + WC (6 %) + AlN, AlB ₂ (4 %) average grain size of cBN is 2 µm								
				TiN/Al ₂ O ₃ /Ti(C, N) coated cemented carbide, mixed ceramic (Al ₂ O ₃ /TiC), and PcBN (90 % CBN and balance with Co)	SNMA 120408 KR 3015 (cemented carbide) SNGA 120408 T 01020 IN 23 (mixed ceramic) SNMA 120408 T 01020 IB 90 (PcBN) VBMT 160408					•		
[170]	2001	Turning	95MnCrW steel (23 HRC)	Multilayer coated carbide (TiN/Al ₂ O ₃ /TiCN)					•	•		
[171]	1999	Milling	22Mn6 steel	Uncoated carbide Fine-grained cemented carbide with different coatings: Ti(C,N), TiAlN+MoS ₂ , (Ti, Al)N, TiN+TiAlN, TiAlOx	–				•			
[172]	2005	Milling	AISI 316 stainless steel (160 HB)	Cemented carbide with different coatings: AlCrN, AlCrNbN, fine-grained (fg)-AlTiN and nano-crystalline (nc)-AlTiN	8-mm diameter finishing end mills				•			
[173]	2010	Milling	H13 (53–55 HRC)	Cemented carbide WC–Co with different coatings: Single-layer nanocrystalline TiAl ₅₅ CrSiYN Multi-layer nanocrystalline TiAlCrSiYN/TiAlCrN	Ball-nose end mills (D = 10 mm)	•	•		•	•		•
[174]	2021	Turning	C20 and C80 steel	TiAlN coated cemented carbide with different coating methods: high power impulse magnetron sputtering (HiPIMS) and direct current magnetron sputtering (DCMS)	–6°, –6°, 6°, 6°, 15°, 75°, 0.8 mm (Orthogonal Rake System – ORS)					•		
[61]	2020	Turning	AISI 304 stainless	Different cemented carbide grades	CNMG120408-MP					•		•

(continued on next page)

Table 7 (continued)

Reference	Year	Process	Workpiece material	Tool material/coated tested	Cutting tool designation/geometry	Output variables evaluated						
						Cutting force and moment	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)	Chip morphology
			steel (201 HB)	Different coatings, coating thickness, and deposition methods: PVD AlTiN and CVD TiN-MT-TiCN- Al_2O_3								
[175]	2008	Milling	H 13 steel (45 HRC)	Uncoated and coated tools (TiN, TiCN, TiAlN, CrN, CrTiAlN)	Two-flute flat micro end mill (D = 500 μm)					•		
[169]	2015	Turning	Hardened steel	Uncoated and coated (TiN, Al_2O_3 and Ti(C, N)) Ti(C,N)-based cermet	TNMG160404	•		•	•	•		•
[176]	2019	Milling	AISI 316L	TiAlN PVD coated WC/Co tool with nanotextured substrate and conventional coated tool	Helix angle of 30°, rake angle of -5°, clearance angle of 5°, and diameter of 10 mm	•	•	•	•	•		

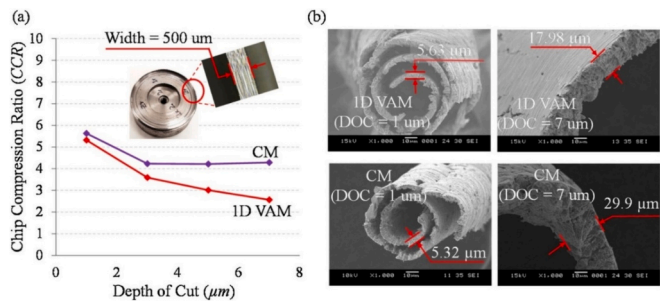


Fig. 11. (a) Chip compression ratio in relation to the depth of cut for the conventional machining (CM) and 1D VAM. (b) SEM images of the chip curling and chip thickness for the CM and the 1D VAM [182]. Workpiece material: SKD11 tool steel. Tool material: CBN (>90 % CBN) 2 μm grit. Cutting parameters tested: $v_c = 0.91\text{--}3.65$ m/min, $v_f = 10$ mm/min, $a_p = 1\text{--}7$ μm .

to the challenge of generalization.

The range of possibilities for the cutting environment is extensive, with a growing inclination towards minimizing the volume of applied oils (MQL) and incorporating nanoparticles into them. These approaches effectively address the environmental concerns associated with extensive oil usage while potentially reducing the cost of machining steels. Nonetheless, the challenge lies in simultaneously maintaining low cutting temperatures, minimizing tool wear, and ensuring the desired quality of the machined components.

Textured tools show promise in improving steel machinability. Research has shown enhancements through textures like grooves, dimples, and chevrons, often created with lasers. Textures serve as lubricant lodgings, trap tool debris, anchor work material, and aid chip movement, reducing friction and improving machining. However, understanding their full impact requires further investigation.

Overcoming these challenges requires continuous research, experimentation, and collaboration between researchers, operators,

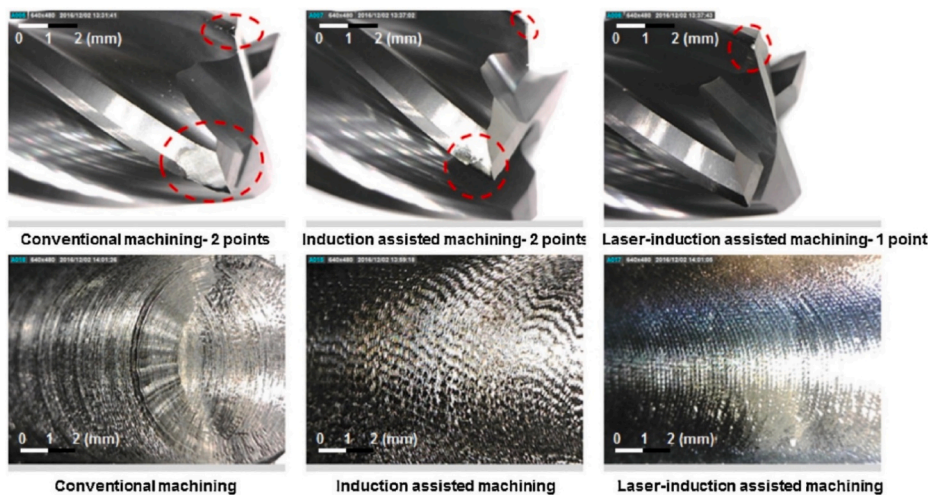


Fig. 12. Comparison between conventional, induction and laser-induction assisted machining. Above the tools used and below the machined surfaces [183]. Workpiece material: AISI 1045. Tool: carbide end mill. $v_c = 125.67$ m/min, $v_f = 120$ mm/min, $a_p = 1.2$ mm.

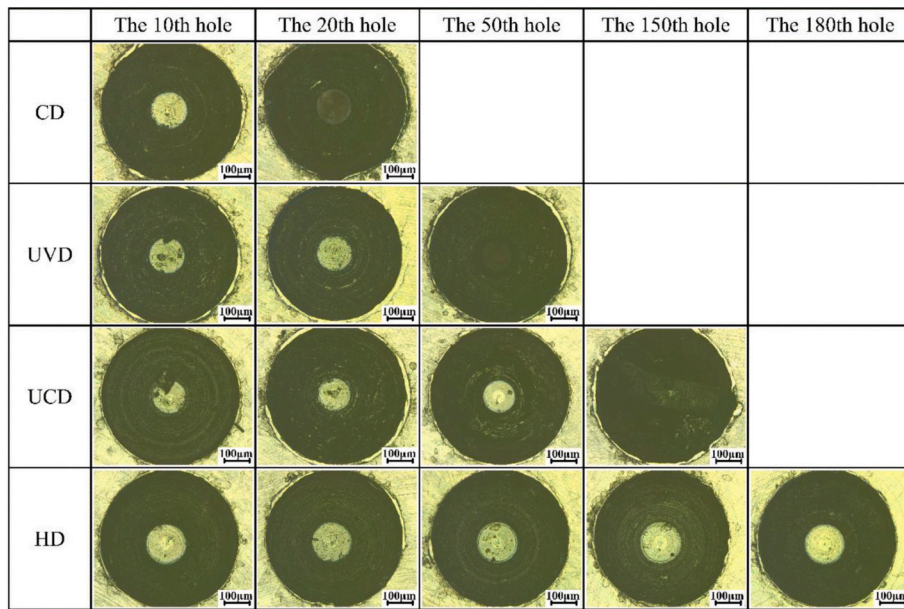


Fig. 13. Micro-hole morphology for the conditions tested by Liang et al. [191]. Workpiece material: AISI 304. Tool: cemented carbide micro-drills ($\varnothing$ 0.5 mm). $v_c = 15.7$ m/min, $v_f = 36$ mm/min, hole depth = 0.8 mm.

industries, and cutting tool manufacturers to successfully machine steels with improved performance and efficiency.

Tool hardness and toughness are two important and antagonistic properties. The former provides tool wear resistance and the latter chipping resistance. Getting the right balance of these two properties in one tool is challenging. Applying artificial intelligence – AI is very much encouraged to find the optimum point. While a coating will generally enhance the machining of steels, finding the ideal coating (including coating type, structure, thickness, number of layers, etc) for each steel grade and machining process also remains a challenging task.

The main challenges of improving the machining of steels through various techniques include optimizing process parameters for vibration-assisted machining (VAM) and ultrasonic-assisted turning (UAT), considering several aspects. Among them, material-specific behavior of different steels, maintaining surface integrity, managing tool wear and tool life, addressing the complexity and cost of implementing these techniques, ensuring compatibility with existing machines, establishing process standardization, and optimizing hybrid machining to achieve synergistic effects. These challenges call for continued research and development to unlock the full potential of these techniques in industrial applications.

With these pointed challenges, the authors hope to encourage researchers all over the world to conduct experimental and numerical studies on the machining of steels, based on the main gaps detected, such as:

- Finding a model that could consider the diverse spectrum of steel grades available, using their mechanical and physical properties, for example, to generalize the machinability of the most important metal on Earth.
- Selecting the ideal tool material and coating.
- Choosing the correct cutting conditions.
- Avoiding the production of long ribbon type of chips that hinders machinability.
- Finding the correct cutting atmosphere not forgetting to look at the sustainable aspects.
- Identifying the ideal assisted method (e.g., Laser Assisted Machining - LAM, Thermally Assisted Machining - TAM, Heat Assisted Machining - HAM) and strategies (e.g., textured tools) to successfully tackle the difficulties of machining steels.

6. Conclusions

This study thoroughly examines steel classification and properties, showcasing its adaptability for diverse applications based on attributes like hardness, machinability, and cost-effectiveness. Common carbon steels, with varying carbon content, cater to different applications: low for ductility and toughness, medium for a balance of strength and ductility, and high for wear resistance and hardness. Alloying enhances these materials for specialized uses.

Stainless steels, distinguished by their chromium-induced corrosion resistance, offer versatility across different applications. Tool steels excel in hardness and wear resistance, ideal for cutting tools, while structural steels are chosen for their strength and weldability in structural applications. The study reveals the intricate relationship between factors impacting steel machinability, highlighting how slight changes in composition, microstructure, and residual elements can notably affect machining performance. Achieving optimal machining parameters requires balancing cutting forces, temperature control, and surface integrity, with techniques like textured tools with MQL showing promise in reducing wear and improving surface finish.

The research indicates that harder steels may elevate cutting forces and temperatures but can yield superior surface finishes with optimal conditions. Nanofluids and tool geometry optimization are crucial for enhancing machinability. Surface integrity, covering surface texture, sub-surface changes, and residual stresses, is vital.

The cutting environment significantly impacts steel machinability. Proper use of cutting fluids can enhance machining performance, enabling higher speeds, feed rates, chip formation, tool life, surface finish, and dimensional accuracy. Different cutting fluid types and application techniques have been assessed for effectiveness.

Dry machining stands out as a cost-effective and eco-friendly option, especially efficient for turning hardened steels with sturdy cutting tools. Lowering cutting fluids' temperature below room temperature enhances their cooling and lubrication, leading to better temperature control and decreased surface roughness during grinding.

High-pressure coolant systems reduce cutting forces, improve chip breakage, extend tool life, and allow for higher cutting speeds and feed rates with less coolant flow. Cryogenic cooling methods surpass dry and MQL techniques by minimizing surface roughness and tool wear through lower cutting zone temperatures, thereby reducing adhesion wear.

Table 8

Improvement of machinability of steels with vibration, thermally and hybrid assisted machining.

Reference	Year	Process	Workpiece material	Tool	Type of assisted machining	Output variables evaluated						
						Cutting force and moment	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)	Chip morphology
[179]	2016	Turning	AISI 4340 hardened steel	CNMA120404	Ultrasonic assisted machining (20 kHz, ~5 μ m amplitude)			•				
[180]	2002	Turning	AISI 4140 Hardened (47 ± 2 HRC) and Annealed (251 HV)	Monocrystalline diamond tool and CBN tool	Vibration-assisted machining Elliptical tool vibration (41 ± 1.5 kHz, 6 μ m amplitude)			•				
[181]	2014	Turning	ASSAB 760 carbon steel	Polycrystalline diamond (PCD) 0.6 mm nose radius with 0° rake angle and 11° clearance angle	Vibration-assisted machining 1D and 2D (38.87 kHz, 2 μ m amplitude)		•					
[182]	2021	Turning	SKD11 tool steel 40 HRC, 50 HRC, and 60 HRC	CBN (>90 % CBN) 2 μ m grit Flat face, clearance angle of 11° , and nose radius of 5 μ m	Vibration-assisted machining Low frequency 1D (0.5 Hz, 10 μ m amplitude) Ultrasonic 1D (38.87 kHz, 10 μ m amplitude)							•
[183]	2018	Milling	AISI 1045	Carbide end mill 8 mm diameter, 100 mm length, 4 flutes	Induction assisted machining and laser-induction assisted machining	•		•		•		
[184]	2016	Milling	AISI 1045	Carbide flat end-mill 6 mm diameter	Laser assisted machining (heat source with 3 mm diameter, 700°C preheating temperature)	•					•	
[185]	2010	Turning	AISI 4130 (50 HRC)	PCBN Nose radius of 1.19 mm, side rake angle of -5° , and back rake angle of 0°	Laser assisted machining: 1 kW Nd:Yag and 1.5 kW CO_2 lasers	•		•				
[186]	2014	Turning	AISI 52100 (54 HRC)	Mixed ceramic SNGA120408-ZNF	Laser assisted machining with different laser mean power (425–560 W), frequency (80–120 Hz), pulse duration, and pulse energy	•		•				
[187]	2015	Milling	17-4PH	CVD coated Ti(C, N)– Al_2O_3 carbide Cutter body diameter: 16 mm XOEX090304FR-ME06 (low feed) LPHW060310TR-MD07 (high feed)	Laser assisted machining Diode laser with different beam size and power	•				•		

(continued on next page)

Table 8 (continued)

Reference	Year	Process	Workpiece material	Tool	Type of assisted machining	Output variables evaluated						
						Cutting force and moment	Temperature	Surface integrity	Tool life	Tool wear	Materials removal rate (MRR)	Chip morphology
[188]	2020	Milling	13-8 stainless steel	TiAlN coated flat end Mill (D = 6 mm)	Laser assisted machining High power diode laser (1 kW) Heat source diameter of 3 mm	•		•				
[190]	2014	Turning	AISI 4140 (50 ± 2 HRC)	CBN and CVD TiN coated cemented tungsten carbide (SNMG 12 04 08)	Plasma enhanced turning (PET) + ultrasonic assisted turning (UAT)	•		•				
[191]	2019	Micro-drilling	AISI 304	Cemented carbide self-fabricated micro-drills (0.5 mm diameter, 0.125 mm web thickness, 1.5 mm flute length, 30° helix angle, 118° point angle, 55° chisel edge angle)	Ultrasonic cavitation and vibration hybrid-assisted machining			•	•	•		•

Advancements in understanding steel machinability have been made, yet the material’s diversity poses significant challenges. Continued research is crucial for enhancing knowledge and refining machining practices. Generalizing machinability across the vast array of steel grades is nearly impossible, emphasizing the importance of characterizing machinability based on material properties, especially ductility and hardness. Variations in chemical content among steel batches may necessitate adjustments in cutting parameters.

Machining of steels has yielded important results that should not be generalized but can inform specific applications. For instance, grooved textured tools with MQL lubricant have been successful in turning AISI 52100 hardened steel; nanoparticles like CuO mixed with deionized water applied via MQL in turning AISI 4340 steel showed better results than other nanoparticles; D2 hardened steel can be machined with mixed ceramic tools to achieve low surface roughness without additional finishing operations.

Cutting lubricants notably improve the surface integrity of steel parts. Metallurgical changes, like phase transformation and grain refinement, often occur during steel machining, particularly under severe cutting conditions or with worn tools. Residual stress, a common sub-surface alteration from machining, impacts fatigue strength and corrosion resistance.

Tool wear, including abrasion, adhesion, diffusion, and oxidation, varies with cutting conditions. Chip control is a major challenge given the diversity of steel types and cutting conditions. Optimizing cutting conditions is vital for process efficiency. Advanced techniques such as Finite Element Analysis (FEA), Discrete Element Method (DEM), Artificial Intelligence (AI), and Genetic Algorithms (GA) can model and optimize these conditions for specific objectives.

The cutting environment plays a pivotal role in steel machining, influencing productivity and component quality. Emerging strategies like Minimum Quantity Lubrication (MQL) and integrating nanoparticles into lubricants show promise for improving machining processes.

The body of research provides industry professionals with a comprehensive guide to enhance their machining strategies. As new materials and technologies emerge, the data from this study will remain crucial in fostering innovation and efficiency in machining processes.

In conclusion, the multitude of factors affecting steel machinability requires a nuanced approach to research and application in industrial settings. The evolving machining technology demands an adaptive methodology considering material properties, tooling innovations, cutting environments, and advanced optimization techniques to meet the demands of modern manufacturing processes.

CRediT authorship contribution statement

Alisson Rocha Machado: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Leonardo Rosa Ribeiro da Silva:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Danil Yu. Pimenov:** Writing – review & editing, Writing – original draft. **Felipe Chagas Rodrigues de Souza:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Mustafa Kuntoğlu:** Writing – review & editing, Writing – original draft. **Raphael Lima de Paiva:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Singh S, Prasad Parida S, Ekka P, Charan Jena P. Characterization of fabricated FG pipe with natural fiber-flyash-epoxy using centrifugal casting. *Int J Innov Technol Explor Eng* 2019;2278–3075. <https://doi.org/10.35940/ijtee.K1437.0981119>.

[2] Parida SP, Jena PC. Free and forced vibration analysis of flyash/graphene filled laminated composite plates using higher order shear deformation theoryvol. 236; 2021. p. 4648–59. <https://doi.org/10.1177/09544062211053181>.

[3] Lo KH, Shek CH, Lai JKL. Recent developments in stainless steels. *Mater Sci Eng R: Rep* 2009;65:39–104. <https://doi.org/10.1016/j.mser.2009.03.001>.

[4] Zhang L, Thomas BG. State of the art in evaluation and control of steel cleanliness. *ISIJ Int* 2003;43:271–91. <https://doi.org/10.2355/isijinternational.43.271>.

- [5] Thewlis G. Materials perspective: classification and quantification of microstructures in steels. *Mater Sci Technol* 2004;20:143–60. <https://doi.org/10.1179/026708304225010325>.
- [6] Suresh R, Basavarajappa S, Gaitonde VN, Samuel GL, Davim JP. State-of-the-art research in machinability of hardened steels. *Proc Inst Mech Eng B J Eng Manuf* 2013;227:191–209. <https://doi.org/10.1177/0954405412464589>.
- [7] Suresh R, Basavarajappa S, Gaitonde VN, Samuel GL, Davim JP. State-of-the-art research in machinability of hardened steels. *Proc Inst Mech Eng B J Eng Manuf* 2013;227:191–209. <https://doi.org/10.1177/0954405412464589>.
- [8] Martinez Krahmer D, Urbicain G, Sánchez Egea AJ. Dry machinability analyses between free cutting, resulfurized, and carbon steels. *Mater Manuf Processes* 2020;35:460–8. <https://doi.org/10.1080/10426914.2020.1734615>.
- [9] Abdulkareem S, Ogedengbe TS, Aweda JO, Ajiboye TK, Khan AA, Babatunde MA. Investigation on effect of material compositions on machinability of carbon steels. *J Phys Conf Ser* 2019;1378:22046. <https://doi.org/10.1088/1742-6596/1378/2/022046> (IOP Publishing).
- [10] Bartarya G, Choudhury SK. State of the art in hard turning. *Int J Mach Tool Manuf* 2012;53:1–14. <https://doi.org/10.1016/j.ijmachtools.2011.08.019>.
- [11] Kaladhar M, Venkata Subbiah K, Srinivasa Rao CH. Machining of austenitic stainless steels - a review. *Int J Mach Mach Mater* 2012;12:178–92. <https://doi.org/10.1504/IJMMM.2012.048564>.
- [12] Krolczyk GM, Maruda RW, Krolczyk JB, Wojciechowski S, Mia M, Nieslony P, et al. Ecological trends in machining as a key factor in sustainable production – a review. *J Clean Prod* 2019;218:601–15. <https://doi.org/10.1016/j.jclepro.2019.02.017>.
- [13] Shaikh MBN, Ali M. Turning of steels under various cooling and lubrication techniques: a review of literature, sustainability aspects, and future scope. *Eng Res Express* 2021;3. <https://doi.org/10.1088/2631-8695/ac2e10>.
- [14] Liao Z, la Monaca A, Murray J, Speidel A, Ushmaev D, Clare A, et al. Surface integrity in metal machining - part I: fundamentals of surface characteristics and formation mechanisms. *Int J Mach Tool Manuf* 2021;162:103687. <https://doi.org/10.1016/j.ijmachtools.2020.103687>.
- [15] Chinchalikar S, Choudhury SK. Machining of hardened steel - experimental investigations, performance modeling and cooling techniques: a review. *Int J Mach Tool Manuf* 2015;89:95–109. <https://doi.org/10.1016/j.ijmachtools.2014.11.002>.
- [16] Goindi GS, Sarkar P. Dry machining: a step towards sustainable machining – challenges and future directions. *J Clean Prod* 2017;165:1557–71. <https://doi.org/10.1016/j.jclepro.2017.07.235>.
- [17] Pan Z, Feng Y, Liang SY. Material microstructure affected machining: a review. *Manuf Rev (Les Ulis)* 2017;4. <https://doi.org/10.1051/mfreview/2017004>.
- [18] Sharif MN, Pervaiz S, Deiab I. Potential of alternative lubrication strategies for metal cutting processes: a review. *Int J Adv Manuf Technol* 2017;89:2447–79. <https://doi.org/10.1007/s00170-016-9298-5>.
- [19] Boswell B, Islam MN, Davies IJ, Ginting YR, Ong AK. A review identifying the effectiveness of minimum quantity lubrication (MQL) during conventional machining. *Int J Adv Manuf Technol* 2017;92:321–40. <https://doi.org/10.1007/s00170-017-0142-3>.
- [20] Ranjan P, Hiremath SS. Role of textured tool in improving machining performance: a review. *J Manuf Process* 2019;43:47–73. <https://doi.org/10.1016/j.jmapro.2019.04.011>.
- [21] Anand A, Behera AK, Das SR. An overview on economic machining of hardened steels by hard turning and its process variables. *Manuf Rev (Les Ulis)* 2019;6. <https://doi.org/10.1051/mfreview/2019002>.
- [22] Sihag N, Sangwan KS. A systematic literature review on machine tool energy consumption. *J Clean Prod* 2020;275. <https://doi.org/10.1016/j.jclepro.2020.123125>.
- [23] Wickramasinghe KC, Sasahara H, Rahim EA, Perera GIP. Green metalworking fluids for sustainable machining applications: a review. *J Clean Prod* 2020;257. <https://doi.org/10.1016/j.jclepro.2020.120552>.
- [24] Singh AK, Kumar A, Sharma V, Kala P. Sustainable techniques in grinding: state of the art review. *J Clean Prod* 2020;269. <https://doi.org/10.1016/j.jclepro.2020.121876>.
- [25] Hamran NNN, Ghani JA, Ramli R, Haron CHC. A review on recent development of minimum quantity lubrication for sustainable machining. *J Clean Prod* 2020;268. <https://doi.org/10.1016/j.jclepro.2020.122165>.
- [26] Yilmaz B, Karabulut Ş, Güllü A. A review of the chip breaking methods for continuous chips in turning. *J Manuf Process* 2020;49:50–69. <https://doi.org/10.1016/j.jmapro.2019.10.026>.
- [27] Sales WF, Schoop J, da Silva LRR, Machado ÁR, Jawahir IS. A review of surface integrity in machining of hardened steels. *J Manuf Process* 2020;58:136–62. <https://doi.org/10.1016/j.jmapro.2020.07.040>.
- [28] Machado AR, da Silva LRR, de Souza FCR, Davis R, Pereira LC, Sales WF, et al. State of the art of tool texturing in machining. *J Mater Process Technol* 2021;293. <https://doi.org/10.1016/j.jmatprotec.2021.117096>.
- [29] Wang B, Liu Z, Cai Y, Luo X, Ma H, Song Q, et al. Advancements in material removal mechanism and surface integrity of high speed metal cutting: a review. *Int J Mach Tool Manuf* 2021;166. <https://doi.org/10.1016/j.ijmachtools.2021.103744>.
- [30] Sankaranarayanan R, RJH N, SK J, Krolczyk GM. A comprehensive review on research developments of vegetable-oil based cutting fluids for sustainable machining challenges. *J Manuf Process* 2021;67:286–313. <https://doi.org/10.1016/j.jmapro.2021.05.002>.
- [31] Chinchalikar S, Kore SS, Hujare P. A review on nanofluids in minimum quantity lubrication machining. *J Manuf Process* 2021;68:56–70. <https://doi.org/10.1016/j.jmapro.2021.05.028>.
- [32] Zadafiya K, Shah P, Shokrani A, Khanna N. Recent advancements in nano-lubrication strategies for machining processes considering their health and environmental impacts. *J Manuf Process* 2021;68:481–511. <https://doi.org/10.1016/j.jmapro.2021.05.056>.
- [33] Ishfaq K, Anjum I, Pruncu CI, Amjad M, Kumar MS, Maqsood MA. Progressing towards sustainable machining of steels: a detailed review. *Materials* 2021;14. <https://doi.org/10.3390/ma14185162>.
- [34] Sultana N, Dhar NR. A critical review on the progress of MQL in machining hardened steels. *Adv Mater Process Technol* 2022;8:3834–58. <https://doi.org/10.1080/2374068X.2022.2036041>.
- [35] Kui GWA, Islam S, Reddy MM, Khandoker N, Chen VLC. Recent progress and evolution of coolant usages in conventional machining methods: a comprehensive review. *Int J Adv Manuf Technol* 2022;119:3–40. <https://doi.org/10.1007/s00170-021-08182-0>.
- [36] Wang Y, Liu C. State-of-the-art on minimum quantity lubrication in green machining. *J Clean Prod* 2023;429. <https://doi.org/10.1016/j.jclepro.2023.139613>.
- [37] Gu G, Wu S, Wang D, Zhou S, Zhu L, An Q, et al. A review of the research on the variation of tool's motion trajectory and its influence on the formation mechanism of surface quality in ultrasonic vibration machining. *J Manuf Process* 2023;107:294–319. <https://doi.org/10.1016/j.jmapro.2023.10.025>.
- [38] El abdelauzi FZ, Jabri A, El Barkany A. Optimization techniques for energy efficiency in machining processes—a review. *Int J Adv Manuf Technol* 2023;125:2967–3001. <https://doi.org/10.1007/s00170-023-10927-y>.
- [39] Cheng Y, Wang Y, Lin J, Xu S, Zhang P. Research status of the influence of machining processes and surface modification technology on the surface integrity of bearing steel materials. *Int J Adv Manuf Technol* 2023;125:2897–923. <https://doi.org/10.1007/s00170-023-10960-x>.
- [40] Liu J, Li Y, Chen Y, Zhou Y, Wang S, Yuan Z, et al. A review of low-temperature plasma-assisted machining: from mechanism to application. *Front Mech Eng* 2023;18. <https://doi.org/10.1007/s11465-022-0734-y>.
- [41] Wang L, Cai W, He Y, Peng T, Xie J, Hu L, et al. Equipment-process-strategy integration for sustainable machining: a review. *Front Mech Eng* 2023;18. <https://doi.org/10.1007/s11465-023-0752-4>.
- [42] Chakraborty S, Datta HN, Chakraborty S. Grey relational analysis-based optimization of machining processes: a comprehensive review. *Proc Integr Optim Sustain* 2023;7:609–39. <https://doi.org/10.1007/s41660-023-00311-4>.
- [43] Hong SY, Ding Y. Micro-temperature manipulation in cryogenic machining of low carbon steelvol. 116; 2001. [https://doi.org/10.1016/S0924-0136\(01\)00836-6](https://doi.org/10.1016/S0924-0136(01)00836-6).
- [44] Kuntoglu M, Sağlam H. Investigation of progressive tool wear for determining of optimized machining parameters in turning. *Measurement* 2019;140:427–36. <https://doi.org/10.1016/j.measurement.2019.04.022>.
- [45] Mia M, Dhar NR. Prediction and optimization by using SVR, RSM and GA in hard turning of tempered AISI 1060 steel under effective cooling condition. *Neural Comput Appl* 2019;31:2349–70. <https://doi.org/10.1007/s00521-017-3192-4>.
- [46] Günay M, Meral T. Modelling and multiresponse optimization for minimizing burr height, thrust force and surface roughness in drilling of ferritic stainless steel. *Sadhana* 2020;45. <https://doi.org/10.1007/s12046-020-01490-3>.
- [47] Zhang W, Wang X, Hu Y, Wang S. Predictive modelling of microstructure changes, micro-hardness and residual stress in machining of 304 austenitic stainless steel. *Int J Mach Tool Manuf* 2018;130–131:36–48. <https://doi.org/10.1016/j.ijmachtools.2018.03.008>.
- [48] Zerti A, Yallesi MA, Zerti O, Nouioua M, Khettabi R. Prediction of machining performance using RSM and ANN models in hard turning of martensitic stainless steel AISI 420. *Proc Inst Mech Eng C J Mech Eng Sci* 2019;233:4439–62. <https://doi.org/10.1177/0954406218820557>.
- [49] Han Y, Liu ZH, Wu CB, Zhao Y, Zu GQ, Zhu WW, et al. A short review on the role of alloying elements in duplex stainless steels. *Tungsten* 2022. <https://doi.org/10.1007/s42864-022-00168-z>.
- [50] Khanna N, Shah P, Maruda RW, Krolczyk GM, Hegab H. Experimental investigation and sustainability assessment to evaluate environmentally clean machining of 15-5 PH stainless steel. *J Manuf Process* 2020;56:1027–38. <https://doi.org/10.1016/j.jmapro.2020.05.016>.
- [51] Sivaiah P, Chakradhar D. Performance improvement of cryogenic turning process during machining of 17-4 PH stainless steel using multi objective optimization techniques. *Measurement (Lond)* 2019;136:326–36. <https://doi.org/10.1016/j.measurement.2018.12.094>.
- [52] Camposco-Negrete C. Prediction and optimization of machining time and surface roughness of AISI O1 tool steel in wire-cut EDM using robust design and desirability approach. *Int J Adv Manuf Technol* 2019;103:2411–22. <https://doi.org/10.1007/s00170-019-03720-3>.
- [53] Mesquita RA, Barbosa CA, Machado AR. Heat treatment of tool steels. In: *Comprehensive materials finishing*. vol. 2–3; 2017. <https://doi.org/10.1016/B978-0-12-803581-8.09191-8>.
- [54] Dossett JL, Boyer HE. *Practical heat treating*. ASM International; 2006.
- [55] Abass MH, Abood AN, Alali M, Hussein SK, Nawi SA. Mechanical properties and microstructure evolution in arc stud welding joints of AISI 1020 with AISI 316L and AISI 304. *Metallogr Microstruct Anal* 2021;10:321–33. <https://doi.org/10.1007/s13632-021-00744-8>.
- [56] Zhao G, Liu L, Wang D, Guo J, Chen W. Mechanical properties of AISI 1045 steel subjected to combined loads of tension and torsion. *Exp Tech* 2018;42:393–406. <https://doi.org/10.1007/s40799-018-0236-3>.
- [57] Mishra A, Maity J. Structure-property correlation of AISI 1080 steel subjected to cyclic quenching treatment. *Mater Sci Eng A* 2015;646:169–81. <https://doi.org/10.1016/j.msea.2015.08.018>.

- [58] Gerin B, Pessard E, Morel F, Verdu C, Mary A. Beneficial effect of prestrain due to cold extrusion on the multiaxial fatigue strength of a 27MnCr5 steel. *Int J Fatigue* 2016;92:345–59. <https://doi.org/10.1016/j.ijfatigue.2016.07.012>.
- [59] Meng L, Lu H, Li W, Guo H, Tian J, Liang W. High strength and plasticity of AISI 430 ferritic stainless steel achieved by a recrystallization annealing before quenching and partitioning process. *Mater Sci Eng A* 2021;814. <https://doi.org/10.1016/j.msea.2021.141191>.
- [60] Becker N, Gilgert J, Petit EJ, Azari Z. The effect of galvanizing on the mechanical resistance and fatigue toughness of a spot welded assembly made of AISI410 martensite. *Mater Sci Eng A* 2014;596:145–56. <https://doi.org/10.1016/j.msea.2013.12.008>.
- [61] He Q, Paiva JM, Kohlscheen J, Beake BD, Veldhuis SC. An integrative approach to coating/carbide substrate design of CVD and PVD coated cutting tools during the machining of austenitic stainless steel. *Ceram Int* 2020;46:5149–58. <https://doi.org/10.1016/j.ceramint.2019.10.259>.
- [62] Rajaguru J, Arunachalam N. Investigation on machining induced surface and subsurface modifications on the stress corrosion crack growth behaviour of super duplex stainless steel. *Corros Sci* 2018;141:230–42. <https://doi.org/10.1016/j.corsci.2018.07.012>.
- [63] Cioacaru EM, Raducanu D, Novicin A, Cinca I, Vintila AN, Serban N, et al. Influence of aging treatment on microstructure and tensile properties of a hot deformed UNS S32750 super duplex stainless steel (SDSS) alloy. *Metals (Basel)* 2020;10. <https://doi.org/10.3390/met10030353>.
- [64] Nong XD, Zhou XL, Li JH, Wang YD, Zhao YF, Brochu M. Selective laser melting and heat treatment of precipitation hardening stainless steel with a refined microstructure and excellent mechanical properties. *Scr Mater* 2020;178:7–12. <https://doi.org/10.1016/j.scriptamat.2019.10.040>.
- [65] Abu Zeid O. On the effect of electrodischarge machining parameters on the fatigue life of AISI D6 tool steel D6 tool steel: 3-D caxit–machining: electrodischarge machining. *J Mater Process Technol* 1997;27 [Elsevier].
- [66] Kuntoglu M. Measurement and analysis of sustainable indicators in machining of Armox 500T armor steel. *Proc Inst Mech Eng C J Mech Eng Sci* 2022. <https://doi.org/10.1177/09544062221079775> [09544062221079775].
- [67] Sivaiah P, Guru Prasad M, Singh MM, Uma B. Machinability evaluation during machining of AISI 52100 steel with textured tools under Minimum Quantity Lubrication—a comparative study. *Mater Manuf Process* 2020;35:1761–8. <https://doi.org/10.1080/10426914.2020.1802034>.
- [68] Pal A, Choudhury SK, Chinchani S. Machinability assessment through experimental investigation during hard and soft turning of hardened steel. *Procedia Mater Sci* 2014;6:80–91. <https://doi.org/10.1016/j.msp.2014.07.010>.
- [69] Das A, Pradhan O, Patel SK, Das SR, Biswal BB. Performance appraisal of various nanofluids during hard machining of AISI 4340 steel. *J Manuf Process* 2019;46:248–70. <https://doi.org/10.1016/j.jmapro.2019.07.023>.
- [70] Shao H, Liu L, Qu HL. Machinability study on 3%Co–12%Cr stainless steel in milling. *Wear* 2007;263:736–44. <https://doi.org/10.1016/j.wear.2007.01.074>.
- [71] Gziesik W. Tribology of metal cutting. *Adv Mach Process Metall Mater* 2008:149–61. <https://doi.org/10.1016/b978-008044534-2.50013-8>.
- [72] Shaw MC. Metal cutting principles vol. 2. New York: Oxford University Press; 1984. [https://doi.org/10.1016/09240136\(91\)90120-4](https://doi.org/10.1016/09240136(91)90120-4).
- [73] Leksyski K, Feldshtein E, Królczuk GM, Legutko S. On the chip shaping and surface topography when finish cutting 17-4 PH precipitation-hardening stainless steel under near-dry cutting conditions. *Materials* 2020;13:2188. <https://doi.org/10.3390/MA13092188>.
- [74] Li X, Liu Z, Liang X. Tool wear, surface topography, and multi-objective optimization of cutting parameters during machining AISI 304 austenitic stainless steel flange. *Metals* 2019;9:972. <https://doi.org/10.3390/MET9090972>.
- [75] Szczotkarz N, Maruda RW, Dębowski D, Leksyski K, Wojciechowski S, Khanna N, et al. Formation of surface topography during turning of AISI 1045 steel considering the type of cutting edge coating. *Adv Sci Technol Res J* 2021;15:253–66. <https://doi.org/10.12913/22998624/141991>.
- [76] Maruda RW, Krolczyk GM, Wojciechowski S, Powalka B, Klos S, Szczotkarz N, et al. Evaluation of turning with different cooling-lubricating techniques in terms of surface integrity and tribologic properties. *Tribol Int* 2020;148. <https://doi.org/10.1016/j.triboint.2020.106334>.
- [77] Mahapatra S, Das A, Jena PC, Das SR. Turning of hardened AISI H13 steel with recently developed S3P-ALTiSiN coated carbide tool using MWCNT mixed nanofluid under minimum quantity lubrication vol. 237; 2022. p. 843–64. <https://doi.org/10.1177/09544062221126357>.
- [78] Zhang W, Wang X, Hu Y, Wang S. Predictive modelling of microstructure changes, micro-hardness and residual stress in machining of 304 austenitic stainless steel. *Int J Mach Tool Manuf* 2018;130–131:36–48. <https://doi.org/10.1016/j.ijmachtools.2018.03.008>.
- [79] Queiroga LR, Marcolino GF, Santos M, Rodrigues G, Eduardo dos Santos C, Brito P. Influence of machining parameters on surface roughness and susceptibility to hydrogen embrittlement of austenitic stainless steels. *Int J Hydrogen Energy* 2019;44:29027–33. <https://doi.org/10.1016/j.ijhydene.2019.09.139>.
- [80] Li B, Zhang S, Hu R, Zhang X. Dislocation density and grain size evolution in hard machining of H13 steel: numerical and experimental investigation. *J Mater Res Technol* 2020;9:4241–54. <https://doi.org/10.1016/j.jmrt.2020.02.051>.
- [81] Li B, Zhang S, Fang Y, Zhang J, Liu Z. Deformation behaviour and texture evolution of martensite steel subjected to hard milling. *Mater Charact* 2019;156:109881. <https://doi.org/10.1016/j.MATCHAR.2019.109881>.
- [82] Zhang F, Duan C, Sun W, Ju K. Effects of cutting conditions on the microstructure and residual stress of white and dark layers in cutting hardened steel. *J Mater Process Technol* 2019;266:599–611. <https://doi.org/10.1016/j.JMATPROTEC.2018.11.038>.
- [83] Clavier F, Valiorgue F, Courbon C, Dumas M, Rech J, van Robaey A, et al. Impact of cutting tool wear on residual stresses induced during turning of a 15-5 PH stainless steel. *Procedia CIRP* 2020;87:107–12. <https://doi.org/10.1016/j.PROCIR.2020.02.074>.
- [84] Da Silva RB, Lima MLS, Pereira MF, Abrão BS, Da Silva LRR, Bianchi EC, et al. A surface and sub-surface quality evaluation of three cast iron grades after grinding under various cutting conditions. *Int J Adv Manuf Technol* 2018;99:1839–52. <https://doi.org/10.1007/s00170-018-2628-z>.
- [85] Zhang W, Li Y, Dong H, Yang C, Jiang X, Lou D, et al. Correlation between machining-induced surface alterations and stress corrosion cracking susceptibility of austenitic stainless steels. *J Mater Res Technol* 2023;26:5076–94. <https://doi.org/10.1016/j.JJMRT.2023.08.239>.
- [86] Zeng H, Hu X, Yang D. Analytical modeling of residual stresses in laser-assisted milling AerMet100 steel. *Opt Laser Technol* 2023;158:108931. <https://doi.org/10.1016/J.OPTLASTEC.2022.108931>.
- [87] Liu Y, Weng J, M'saoubi R, Ahadi A, Zhou J. Evolution of residual stress when turning a fillet radius in stainless steel. *J Manuf Process* 2023;85:216–26. <https://doi.org/10.1016/J.JMAPRO.2022.11.038>.
- [88] Yang S, Jin X, Engin S, Kountanya R, El-Wardany T, Lee SY. Effect of cutting fluids on surface residual stress in machining of wasp alloy. *J Mater Process Technol* 2023;322:118170. <https://doi.org/10.1016/J.JMATPROTEC.2023.118170>.
- [89] Damir A, Sadek A, Attia H. On the performance of cryogenic technology in milling of hardened H13 tool steel. *Procedia CIRP* 2024;122:384–9. <https://doi.org/10.1016/J.PROCIR.2024.02.012>.
- [90] Barcelos MB, de Almeida DT, Tusset F, Scheuer CJ. Performance analysis of conventional and high-feed turning tools in machining the thermally affected zone after plasma arc cutting of low carbon manganese-alloyed steel. *J Manuf Process* 2024;115:18–39. <https://doi.org/10.1016/J.JMAPRO.2024.01.088>.
- [91] Zhou T, Zhou T, Zhang C, Sun C, Cui H, Tian P, et al. Hybrid modeling with finite element—analysis—neural network for predicting residual stress in orthogonal cutting of H13. *J Mater Res Technol* 2024;29:4954–77. <https://doi.org/10.1016/J.JJMRT.2024.02.126>.
- [92] Sun J, Dilger K. Influence of initial residual stresses on welding residual stresses in ultra-high strength steel S960. *J Manuf Process* 2023;101:259–68. <https://doi.org/10.1016/J.JMAPRO.2023.06.011>.
- [93] Liu S, Wang X, Liu Z, Chen H, Wang P. Collaborative formation pattern of surface topography and residual stress in hard turning of steels. *J Mater Res Technol* 2023;26:737–52. <https://doi.org/10.1016/J.JJMRT.2023.07.217>.
- [94] Sreejith S, Priyadarshini A, Chaganti PK, Prabhu G, Mylavaram P. Effect of machining operations on mechanical properties, surface integrity and corrosion resistance of tungsten heavy alloy. *Mater Today Commun* 2023;37:106930. <https://doi.org/10.1016/J.MTCOMM.2023.106930>.
- [95] Xu C, Wang X, Geng Y, Wang Y, Sun Z, Yu B, et al. Effect of shot peening on the surface integrity and fatigue property of gear steel 16Cr3NiWMoVnBe at room temperature. *Int J Fatigue* 2023;172:107668. <https://doi.org/10.1016/J.IJFATIGUE.2023.107668>.
- [96] Khatai S, Kumar Sahoo A, Kumar R, Panda A. On machining behaviour of various cutting inserts: a review on hardened steel. *Mater Today Proc* 2022;62:3485–92. <https://doi.org/10.1016/j.matpr.2022.04.301>.
- [97] Zhao J, Liu Z. Influences of coating thickness on cutting temperature for dry hard turning Inconel 718 with PVD TiAlN coated carbide tools in initial tool wear stage. *J Manuf Process* 2020;56:1155–65. <https://doi.org/10.1016/j.jmapro.2020.06.010>.
- [98] Jamil M, He N, Gupta MK, Zhao W, Khan AM. Tool wear mechanisms and its influence on machining tribology of face milled titanium alloy under sustainable hybrid lubri-cooling. *Tribol Int* 2022;170. <https://doi.org/10.1016/j.triboint.2022.107497>.
- [99] Gupta MK, el Etri H, Korkmaz ME, Ross NS, Krolczyk GM, Gawlik J, et al. Tribological and surface morphological characteristics of titanium alloys: a review. *Arch Civil Mech Eng* 2022;22:1–23. <https://doi.org/10.1007/s43452-022-00392-x>.
- [100] Jamil M, He N, Huang X, Zhao W, Gupta MK, Khan AM. Measurement of machining characteristics under novel dry ice blasting cooling assisted milling of AISI 52100 tool steel. *Measurement (Lond)* 2022;191. <https://doi.org/10.1016/j.measurement.2022.110821>.
- [101] Colantonio L, Equeter L, Dehombreux P, Ducobu F. A systematic literature review of cutting tool wear monitoring in turning by using artificial intelligence techniques. *Machines* 2021;9. <https://doi.org/10.3390/machines9120351>.
- [102] Aslan A. Optimization and analysis of process parameters for flank wear, cutting forces and vibration in turning of AISI 5140: a comprehensive study. *Measurement (Lond)* 2020;163:107959. <https://doi.org/10.1016/j.measurement.2020.107959>.
- [103] Hatt O, Crawford P, Jackson M. On the mechanism of tool crater wear during titanium alloy machining. *Wear* 2017;374–375:15–20. <https://doi.org/10.1016/j.wear.2016.12.036>.
- [104] Kamble PD, Waghmare AC, Askhedkar RD, Sahare SB. Multi objective optimization of turning parameters considering spindle vibration by Hybrid Taguchi Principal component analysis (HTPCA). *Mater Today Proc* 2017;4:2077–84. <https://doi.org/10.1016/j.matpr.2017.02.053>.
- [105] Liao Z, Abdelhafeez A, Li H, Yang Y, Diaz OG, Axinte D. State-of-the-art of surface integrity in machining of metal matrix composites. *Int J Mach Tool Manuf* 2019;143:63–91. <https://doi.org/10.1016/j.ijmachtools.2019.05.006>.

- [106] Pacella M. A new low-feed chip breaking tool and its effect on chip morphology. *Int J Adv Manuf Technol* 2019;104:1145–57. <https://doi.org/10.1007/s00170-019-03961-2>.
- [107] Zhu Z, Guo X, Ekevad M, Cao P, Na B, Zhu N. The effects of cutting parameters and tool geometry on cutting forces and tool wear in milling high-density fiberboard with ceramic cutting tools. *Int J Adv Manuf Technol* 2017;91:4033–41. <https://doi.org/10.1007/s00170-017-0085-8>.
- [108] Sales WF, Schoop J, da Silva LRR, Machado AR, Jawahir IS. A review of surface integrity in machining of hardened steels. *J Manuf Process* 2020;58:136–62. <https://doi.org/10.1016/j.jmapro.2020.07.040>.
- [109] Usca UA, Uzun M, Kuntoğlu M, Sap E, Gupta MK. Investigations on tool wear, surface roughness, cutting temperature, and chip formation in machining of Cu-B-CrC composites. *Int J Adv Manuf Technol* 2021;116:3011–25. <https://doi.org/10.1007/s00170-021-07670-7>.
- [110] Sap E, Usca UA, Gupta MK, Kuntoğlu M. Tool wear and machinability investigations in dry turning of Cu/Mo-SiCp hybrid composites. *Int J Adv Manuf Technol* 2021;114:379–96. <https://doi.org/10.1007/s00170-021-06889-8>.
- [111] Han F, Li L, Cai W, Li C, Deng X, Sutherland JW. Parameters optimization considering the trade-off between cutting power and MRR based on Linear Decreasing Particle Swarm Algorithm in milling. *J Clean Prod* 2020;262. <https://doi.org/10.1016/j.jclepro.2020.121388>.
- [112] Zerti O, Yallese MA, Khettabi R, Chaoui K, Mabrouki T. Design optimization for minimum technological parameters when dry turning of AISI D3 steel using Taguchi method. *Int J Adv Manuf Technol* 2017;89:1915–34. <https://doi.org/10.1007/s00170-016-9162-7>.
- [113] Xu L, Huang C, Li C, Wang J, Liu H, Wang X. A novel intelligent reasoning system to estimate energy consumption and optimize cutting parameters toward sustainable machining. *J Clean Prod* 2020;261:121160. <https://doi.org/10.1016/j.jclepro.2020.121160>.
- [114] Zhou G, Lu Q, Xiao Z, Zhou C, Tian C. Cutting parameter optimization for machining operations considering carbon emissions. *J Clean Prod* 2019;208:937–50. <https://doi.org/10.1016/j.jclepro.2018.10.191>.
- [115] Alok A, Das M. Multi-objective optimization of cutting parameters during sustainable dry hard turning of AISI 52100 steel with newly developed HSN2-coated carbide insert. *Measurement (Lond)* 2019;133:288–302. <https://doi.org/10.1016/j.measurement.2018.10.009>.
- [116] Singh BK, Roy H, Mondal B, Roy SS, Mandal N. Measurement of chip morphology and multi criteria optimization of turning parameters for machining of AISI 4340 steel using Y-ZTA cutting insert. *Measurement (Lond)* 2019;142:181–94. <https://doi.org/10.1016/j.measurement.2019.04.064>.
- [117] Anandan V, Naresh Babu M, Vetrivel Sezhian M, Yildirim CV, Dinesh Babu M. Influence of graphene nanofluid on various environmental factors during turning of M42 steel. *J Manuf Process* 2021;68:90–103. <https://doi.org/10.1016/j.jmapro.2021.07.019>.
- [118] Barzegar Z, Ozlu E. Analytical prediction of cutting tool temperature distribution in orthogonal cutting including third deformation zone. *J Manuf Process* 2021;67:325–44. <https://doi.org/10.1016/j.jmapro.2021.05.003>.
- [119] Saez-de-Buruaga M, Soler D, Aristimuño PX, Esnaola JA, Arrazola PJ. Determining tool/chip temperatures from thermography measurements in metal cutting. *Appl Therm Eng* 2018;145:305–14. <https://doi.org/10.1016/j.applthermaleng.2018.09.051>.
- [120] Rajaguru J, Dwivedi M, Natarajan S, Krishnaswamy H, Arunachalam N. Machining induced residual stress prediction during orthogonal cutting of super duplex stainless steel using CEL approach. *J Manuf Process* 2022;82:474–87. <https://doi.org/10.1016/j.jmapro.2022.07.071>.
- [121] Wang J, Zhang J, Peng R, Zhao X, Zhao G, Chen N, et al. Research on cutting parameters of low-temperature liquid CO₂ assisted PCD turning bearing ring. *J Manuf Process* 2023;87:199–208. <https://doi.org/10.1016/j.jmapro.2023.01.009>.
- [122] Li G, Xu W, Jin X, Liu L, Ding S, Li C. The machinability of stainless steel 316 L fabricated by selective laser melting: typical cutting responses, white layer and evolution of chip morphology. *J Mater Process Technol* 2023;315. <https://doi.org/10.1016/j.jmatprotec.2023.117926>.
- [123] Wu S, He Y, Zhu T, Zhang W, Chen W, Wang C. Microstructure and mechanical properties of superficial surface and subsurface layers in the cutting of hardened steel under cryogenic cooling. *J Mater Process Technol* 2023;322. <https://doi.org/10.1016/j.jmatprotec.2023.118165>.
- [124] Li C, Chen X, Tang Y, Li L. Selection of optimum parameters in multi-pass face milling for maximum energy efficiency and minimum production cost. *J Clean Prod* 2017;140:1805–18. <https://doi.org/10.1016/j.jclepro.2016.07.086>.
- [125] Li B, Zhang S, Zhang Q, Chen J, Zhang J. Modelling of phase transformations induced by thermo-mechanical loads considering stress-strain effects in hard milling of AISI H13 steel. *Int J Mech Sci* 2018;149:241–53. <https://doi.org/10.1016/j.jimecs.2018.10.010>.
- [126] Lopes JG, Machado CM, Duarte VR, Rodrigues TA, Santos TG, Oliveira JP. Effect of milling parameters on HSLA steel parts produced by Wire and Arc Additive Manufacturing (WAAM). *J Manuf Process* 2020;59:739–49. <https://doi.org/10.1016/j.jmapro.2020.10.007>.
- [127] Mokas N, Boulanour L, Amirat A, Gautier L. Effect of hardening on the machinability of C22 steel under drilling operations using twist HSS drill bit. *Int J Adv Manuf Technol* 2018;96:3227–34. <https://doi.org/10.1007/s00170-018-1818-z>.
- [128] Winter M, Li W, Kara S, Herrmann C. Determining optimal process parameters to increase the eco-efficiency of grinding processes. *J Clean Prod* 2014;66:644–54. <https://doi.org/10.1016/j.jclepro.2013.10.031>.
- [129] Deng Z, Lv L, Li S, Wan L, Liu W, Yan C, et al. Study on the model of high efficiency and low carbon for grinding parameters optimization and its application. *J Clean Prod* 2016;137:1672–81. <https://doi.org/10.1016/j.jclepro.2016.03.022>.
- [130] Zhou W, Tang J, Shao W. Study on surface generation mechanism and roughness distribution in gear profile grinding. *Int J Mech Sci* 2020;187:105921. <https://doi.org/10.1016/j.jimecs.2020.105921>.
- [131] Khare SK, Agarwal S. Predictive modeling of surface roughness in grinding. *Procedia CIRP* 2015;31:375–80. <https://doi.org/10.1016/j.procir.2015.04.092>.
- [132] Stephenson DA, Agapiou JS. *Metal cutting theory and practice*. 3rd ed. Boca Raton, FL: CRC Press; 2016.
- [133] Sarikaya M, Gupta MK, Tomaz I, Danish M, Mia M, Rubaiee S, et al. Cooling techniques to improve the machinability and sustainability of light-weight alloys: a state-of-the-art review. *J Manuf Process* 2021;62:179–201. <https://doi.org/10.1016/j.jmapro.2020.12.013>.
- [134] Liew PJ, Shaaroni A, Sidik NAC, Yan J. An overview of current status of cutting fluids and cooling techniques of turning hard steel. *Int J Heat Mass Transf* 2017;114:380–94. <https://doi.org/10.1016/j.jheatmasstransfer.2017.06.077>.
- [135] Najiha MS, Rahman MM, Yusoff AR. Environmental impacts and hazards associated with metal working fluids and recent advances in the sustainable systems: a review. *Renew Sustain Energy Rev* 2016;60:1008–31. <https://doi.org/10.1016/j.rser.2016.01.065>.
- [136] Goindi GS, Sarkar P. Dry machining: a step towards sustainable machining – challenges and future directions. *J Clean Prod* 2017;165:1557–71. <https://doi.org/10.1016/j.jclepro.2017.07.235>.
- [137] de Paiva RL, da Silva RB, Jackson MJ, Abrão AM. The influence of cutting fluid concentration on surface integrity of VP80 steel and the influence of cutting fluid flow rate on surface roughness of VPATLAS steel after grinding. *J Manuf Sci Eng* 2017;139:1–7. <https://doi.org/10.1115/1.4038149>.
- [138] de Paiva RL, de Souza Ruzzi R, da Silva RB. An approach to reduce thermal damages on grinding of bearing steel by controlling cutting fluid temperature. *Metals (Basel)* 2021;11:1660. <https://doi.org/10.3390/met11101660>.
- [139] Araújo RP, Rolim TL, Oliveira CA, Moura AE, Silva JCA. Analysis of the surface roughness and cutting tool wear using a vapor compression assisted cooling system to cool the cutting fluid in turning operation. *J Manuf Process* 2019;44:38–46. <https://doi.org/10.1016/j.jmapro.2019.05.040>.
- [140] Kramar D, Krajnik P, Kopac J. Capability of high pressure cooling in the turning of surface hardened piston rods. *J Mater Process Technol* 2010;210:212–8. <https://doi.org/10.1016/j.jmatprotec.2009.09.002>.
- [141] Mia M, Dhar NR. Optimization of surface roughness and cutting temperature in high-pressure coolant-assisted hard turning using Taguchi method. *Int J Adv Manuf Technol* 2017;88:739–53. <https://doi.org/10.1007/s00170-016-8810-2>.
- [142] Sredanović B, Globočki Lakić G. Hard turning of bearing steel AISI 52100 with carbide tool and high pressure coolant supply. *J Braz Soc Mech Sci Eng* 2017;39:4623–32. <https://doi.org/10.1007/s40430-017-0764-2>.
- [143] Khanna N, Shah P, Chetan.. Comparative analysis of dry, flood, MQL and cryogenic CO₂ techniques during the machining of 15-5-PH SS alloy. *Tribol Int* 2020;146. <https://doi.org/10.1016/j.triboint.2020.106196>.
- [144] Sterle L, Krajnik P, Pušavec F. The effects of liquid-CO₂ cooling, MQL and cutting parameters on drilling performance. *CIRP Ann* 2021;70:79–82. <https://doi.org/10.1016/j.cirp.2021.04.007>.
- [145] Lai Z, Wang C, Zheng L, Huang W, Yang J, Guo G, et al. Adaptability of AlTiN-based coated tools with green cutting technologies in sustainable machining of 316L stainless steel. *Tribol Int* 2020;148. <https://doi.org/10.1016/j.triboint.2020.106300>.
- [146] Sato BK, Lopes JC, Diniz AE, Rodrigues AR, de Mello HJ, Sanchez LEA, et al. Toward sustainable grinding using minimum quantity lubrication technique with diluted oil and simultaneous wheel cleaning. *Tribol Int* 2020;147. <https://doi.org/10.1016/j.triboint.2020.106276>.
- [147] Lima de Paiva R, de Souza Ruzzi R, Batista da Silva R. Contribution to the selection of cutting fluid type and its application technique for grinding of bearing steel. *Proc Inst Mech Eng B J Eng Manuf* 2021;1–11. <https://doi.org/10.1177/09544054211040633>.
- [148] Rajaguru J, Arunachalam N. A comprehensive investigation on the effect of flood and MQL coolant on the machinability and stress corrosion cracking of super duplex stainless steel. *J Mater Process Technol* 2020;276:116417. <https://doi.org/10.1016/j.jmatprotec.2019.116417>.
- [149] Muaz M, Choudhury SK. Experimental investigations and multi-objective optimization of MQL-assisted milling process for finishing of AISI 4340 steel. *Measurement (Lond)* 2019;138:557–69. <https://doi.org/10.1016/j.measurement.2019.02.048>.
- [150] Pal A, Chatha SS, Sidhu HS. Experimental investigation on the performance of MQL drilling of AISI 321 stainless steel using nano-graphene enhanced vegetable-oil-based cutting fluid. *Tribol Int* 2020;151. <https://doi.org/10.1016/j.triboint.2020.106508>.
- [151] Anandan V, Naresh Babu M, Vetrivel Sezhian M, Yildirim CV, Dinesh Babu M. Influence of graphene nanofluid on various environmental factors during turning of M42 steel. *J Manuf Process* 2021;68:90–103. <https://doi.org/10.1016/j.jmapro.2021.07.019>.
- [152] de Paiva RL, de Souza Ruzzi R, de Oliveira LR, Bandarra Filho EP, Gonçalves Neto LM, Gelamo RV, et al. Experimental study of the influence of graphene platelets on the performance of grinding of SAE 52100 steel. *Int J Adv Manuf Technol* 2020;110. <https://doi.org/10.1007/s00170-020-05866-x>.
- [153] de Paiva RL, de Oliveira D, Ruzzi R de S, Jackson MJ, Gelamo RV, Abrão AM, et al. Surface roughness and morphology evaluation of bearing steel after grinding

- with multilayer graphene platelets dispersed in different base fluids. *Wear* 2023; 523. <https://doi.org/10.1016/j.wear.2023.204852>.
- [154] Peña-Parás L, Maldonado-Cortés D, Rodríguez-Villalobos M, Romero-Cantú AG, Montemayor OE, Herrera M, et al. Optimization of milling parameters of 1018 steel and nanoparticle additive concentration in cutting fluids for enhancing multi-response characteristics. *Wear* 2019;426–427:877–86. <https://doi.org/10.1016/j.wear.2019.01.078>.
- [155] Dogra M, Sharma VS, Dureja J. Effect of tool geometry variation on finish turning - a review vol. 4; 2011. <https://doi.org/10.25103/jestr.041.01>.
- [156] Aramcharoen A, Mativenga PT. Size effect and tool geometry in micromilling of tool steel. *Precis Eng* 2009;33:402–7. <https://doi.org/10.1016/j.precisioneng.2008.11.002>.
- [157] Machado AR, da Silva LRR, de Souza FCR, Davis R, Pereira LC, Sales WF, et al. State of the art of tool texturing in machining. *J Mater Process Technol* 2021;293: 117096. <https://doi.org/10.1016/j.jmatprotec.2021.117096>.
- [158] Ma J, Duong NH, Lei S. Numerical investigation of the performance of microbump textured cutting tool in dry machining of AISI 1045 steel. *J Manuf Process* 2015; 19:194–204. <https://doi.org/10.1016/j.jmapro.2014.10.001>.
- [159] Duan R, Deng J, Ai X, Liu Y, Chen H. Experimental assessment of derivative cutting of micro-textured tools in dry cutting of medium carbon steels. *Int J Adv Manuf Technol* 2017;92:3531–40. <https://doi.org/10.1007/s00170-017-0360-8>.
- [160] Sivaiah P. Evaluation of hybrid textured tool performance under minimum quantity lubrication while turning of AISI 304 steel. *J Braz Soc Mech Sci Eng* 2019;41. <https://doi.org/10.1007/s40430-019-2069-0>.
- [161] Vasumathy D, Meena A. Influence of micro scale textured tools on tribological properties at tool-chip interface in turning AISI 316 austenitic stainless steel. *Wear* 2017;376–377:1747–58. <https://doi.org/10.1016/j.wear.2017.01.024>.
- [162] Zhang K, Deng J, Sun J, Jiang C, Liu Y, Chen S. Effect of micro/nano-scale textures on anti-adhesive wear properties of WC/Co-based TiAlN coated tools in AISI 316 austenitic stainless steel cutting. *Appl Surf Sci* 2015;355:602–14. <https://doi.org/10.1016/j.apsusc.2015.07.132>.
- [163] Rajbongshi SK, Annebushan Singh M, Kumar Sarma D. A comparative study in machining of AISI D2 steel using textured and non-textured coated carbide tool at the flank face. *J Manuf Process* 2018;36:360–72. <https://doi.org/10.1016/j.jmapro.2018.10.041>.
- [164] Duan R, Deng J, Lei S, Ge D, Liu Y, Li X. Effect of derivative cutting on machining performance of micro textured tools. *J Manuf Process* 2019;45:544–56. <https://doi.org/10.1016/j.jmapro.2019.07.037>.
- [165] Law M, Karthik R, Sharma S, Ramkumar J. Finish turning of hardened bearing steel using textured PcBN tools. *J Manuf Process* 2020;60:144–61. <https://doi.org/10.1016/j.jmapro.2020.10.051>.
- [166] Xing Y, Deng J, Zhao J, Zhang G, Zhang K. Cutting performance and wear mechanism of nanoscale and microscale textured $\text{Al}_2\text{O}_3/\text{TiC}$ ceramic tools in dry cutting of hardened steel. *Int J Refract Met Hard Mater* 2014;43:46–58. <https://doi.org/10.1016/j.jirmhm.2013.10.019>.
- [167] Sobiyi K, Sigalas I, Akdogan G, Turan Y. Performance of mixed ceramics and CBN tools during hard turning of martensitic stainless steel. *Int J Adv Manuf Technol* 2015;77:861–71. <https://doi.org/10.1007/s00170-014-6506-z>.
- [168] Sales WF, Costa LA, Santos SC, Diniz AE, Bonney J, Ezugwu EO. Performance of coated, cemented carbide, mixed-ceramic and PCBN-H tools when turning W320 steel. *Int J Adv Manuf Technol* 2009;41:660–9. <https://doi.org/10.1007/s00170-008-1523-4>.
- [169] Chen X, Xu J, Xiao Q. Cutting performance and wear characteristics of Ti(C,N)-based cermet tool in machining hardened steel. *Int J Refract Met Hard Mater* 2015;52:143–50. <https://doi.org/10.1016/j.jirmhm.2015.06.006>.
- [170] Che Haron CH, Ginting A, Goh JH. Wear of coated and uncoated carbides in turning tool steel vol. 116; 2001. [https://doi.org/10.1016/S0924-0136\(01\)00841-X](https://doi.org/10.1016/S0924-0136(01)00841-X).
- [171] Lahres M, Doerfel O, Neumüller R. Applicability of different hard coatings in dry machining an austenitic steel vol. 120–121; 1999. [https://doi.org/10.1016/S0257-8972\(99\)00359-X](https://doi.org/10.1016/S0257-8972(99)00359-X).
- [172] Endrino JL, Fox-Rabinovich GS, Gey C. Hard AlTiN, AlCrN PVD coatings for machining of austenitic stainless steel. *Surf Coat Technol* 2006;200:6840–5. <https://doi.org/10.1016/j.surfcoat.2005.10.030>.
- [173] Fox-Rabinovich GS, Yamamoto K, Beake BD, Kovalev AI, Aguirre MH, Veldhuis SC, et al. Emergent behavior of nano-multilayered coatings during dry high-speed machining of hardened tool steels. *Surf Coat Technol* 2010;204: 3425–35. <https://doi.org/10.1016/j.surfcoat.2010.04.002>.
- [174] Kumar A, Bauri R, Naskar A, Chattopadhyay AK. Characterization of HiPIMS and DCMS deposited TiAlN coatings and machining performance evaluation in high speed dry machining of low and high carbon steel. *Surf Coat Technol* 2021;417. <https://doi.org/10.1016/j.surfcoat.2021.127180>.
- [175] Aramcharoen A, Mativenga PT, Yang S, Cooke KE, Teer DG. Evaluation and selection of hard coatings for micro milling of hardened tool steel. *Int J Mach Tool Manuf* 2008;48:1578–84. <https://doi.org/10.1016/j.ijmachtools.2008.05.011>.
- [176] Zhang K, Guo X, Sun L, Meng X, Xing Y. Fabrication of coated tool with femtosecond laser pretreatment and its cutting performance in dry machining SLM-produced stainless steel. *J Manuf Process* 2019;42:28–40. <https://doi.org/10.1016/j.jmapro.2019.04.009>.
- [177] Zhou M, Eow YT, Ngoi BKA, Lim EN. Vibration-assisted precision machining of steel with PCD tools. *Mater Manuf Process* 2003;18:825–34. <https://doi.org/10.1081/AMP-120024978>.
- [178] Yang Z, Zhu L, Zhang G, Ni C, Lin B. Review of ultrasonic vibration-assisted machining in advanced materials. *Int J Mach Tool Manuf* 2020;156:103594. <https://doi.org/10.1016/j.ijmachtools.2020.103594>.
- [179] Sharma V, Pandey PM. Optimization of machining and vibration parameters for residual stresses minimization in ultrasonic assisted turning of 4340 hardened steel. *Ultrasonics* 2016;70:172–82. <https://doi.org/10.1016/j.ultras.2016.05.001>.
- [180] Willert M, Zielinski T, Rickens K, Riemer O, Karpuschewski B. Impact of ultrasonic assisted cutting of steel on surface integrity. *Procedia CIRP* 2020;vol. 87:222–7. <https://doi.org/10.1016/j.procir.2020.02.009> [Elsevier B.V.].
- [181] Zhang X, Liu K, Kumar AS, Rahman M. A study of the diamond tool wear suppression mechanism in vibration-assisted machining of steel. *J Mater Process Technol* 2014;214:496–506. <https://doi.org/10.1016/j.jmatprotec.2013.10.002>.
- [182] Arefin S, Zhang XQ, Kumar AS, Neo DWK, Wang Y. Study of chip formation mechanism in one-dimensional vibration-assisted machining. *J Mater Process Technol* 2021;291. <https://doi.org/10.1016/j.jmatprotec.2020.117022>.
- [183] Baek JT, Woo WS, Lee CM. A study on the machining characteristics of induction and laser-induction assisted machining of AISI 1045 steel and Inconel 718. *J Manuf Process* 2018;34:513–22. <https://doi.org/10.1016/j.jmapro.2018.06.030>.
- [184] Ahn JW, Woo WS, Lee CM. A study on the energy efficiency of specific cutting energy in laser-assisted machining. *Appl Therm Eng* 2016;94:748–53. <https://doi.org/10.1016/j.applthermaleng.2015.10.129>.
- [185] Ding H, Shin YC. Laser-assisted machining of hardened steel parts with surface integrity analysis. *Int J Mach Tool Manuf* 2010;50:106–14. <https://doi.org/10.1016/j.ijmachtools.2009.09.001>.
- [186] Panjehpour A, Soleymani Yazdi MR, Shoja-Razavi R. An experimental investigation of pulsed laser-assisted machining of AISI 52100 steel. *Opt Laser Technol* 2014;63:137–43. <https://doi.org/10.1016/j.optlastec.2014.03.018>.
- [187] Birmingham MJ, Kent D, Dargusch MS. A new understanding of the wear processes during laser assisted milling 17-4 precipitation hardened stainless steel. *Wear* 2015;328–329:518–30. <https://doi.org/10.1016/j.wear.2015.03.025>.
- [188] Cao XF, Woo WS, Lee CM. A study on the laser-assisted milling of 13-8 stainless steel for optimal machining. *Opt Laser Technol* 2020;132. <https://doi.org/10.1016/j.optlastec.2020.106473>.
- [189] Lauwers B. Surface integrity in hybrid machining processes. *Procedia Eng* 2011; 19:241–51. <https://doi.org/10.1016/j.proeng.2011.11.107> [Elsevier Ltd].
- [190] Farahnakian M, Razfar MR. Experimental study on hybrid ultrasonic and plasma aided turning of hardened steel AISI 4140. *Mater Manuf Process* 2014;29:550–6. <https://doi.org/10.1080/10426914.2014.892612>.
- [191] Liang Z, Ma Y, Nie Q, Wang X, Zhou T, Guo H, et al. Ultrasonic cavitation and vibration hybrid-assisted micro-drilling of stainless steel. *Int J Adv Manuf Technol* 2019;104:3073–82. <https://doi.org/10.1007/s00170-019-04247-3>.